

Continuous Heterogeneity Estimation for Self-Selection (CHESS): Evaluating the Persuasive Effects of Issue Ads

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Abstract

The persuasive effects of political messages depend on whether audiences choose to engage with them. However, experiments testing political messages typically force participants to view content, mechanically preventing self-selection and producing estimates that may systematically differ from those realized under voluntary viewing. I introduce a new experimental design, Continuous Heterogeneity Estimation for Self-Selection (CHESS), that recovers how persuasive effects vary across individuals with different selection propensities. In contrast to binary choice designs such as PICA, CHESS recovers continuous heterogeneous treatment effects while improving efficiency and avoiding sensitivity to the researcher’s choice of control condition. In two preregistered experiments, I find that a pro-gun-control ad persuades high-propensity viewers while an anti-gun-control ad persuades low-propensity viewers. The anti-gun-control ad reduces the priority of gun laws only among medium-propensity viewers. These results illustrate how CHESS enables researchers to uncover fine-grained, potentially nonlinear persuasive patterns that average treatment effects obscure.

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Introduction

Experiments now play a key role in guiding political messaging decisions (Hewitt et al., 2024). For example, during the 2024 US presidential race, the Democratic super PAC Future Forward tested over 4,000 Harris ads to inform its \$700 million advertising effort, with the chief data scientist explaining that “for any given ad, you take a thousand people, split them into treatment and control, and then compare outcomes” (Klein, 2025; Schleifer and Goldmacher, 2024). This standard experimental design recovers the average treatment effect (ATE) of exposure to a campaign ad. However, in practice, exposure to ads involves self-selection: many viewers tune out when ads come on. From a campaign’s perspective, the relevant question is not “How persuasive is this ad to a random sample of the population?”, but rather “How persuasive is this ad to the people who are likely to sit through it?”

This example highlights a well-known methodological problem: in many settings, people must opt into interventions. In the domain of political communication, people choose whether or not to engage with ideologically incongruent news, open the door for a canvasser, or attend a political rally. More generally, people opt into government benefit programs or medical treatments. Across these contexts, standard experiments estimate average treatment effects, but researchers and practitioners may care about heterogeneity by the propensity to self-select into treatment: who is likely to opt into an intervention, and how effective is the intervention for those people?

I introduce a new experimental design—Continuous Heterogeneity Estimation for Self-Selection (CHESS)—that allows researchers to answer these kinds of questions, and I apply this design to estimate the efficacy of gun-control issue ads. The main idea behind CHESS is to directly elicit a measure of participants’ willingness to self-select into treatment using the Becker–DeGroot–Marschak (BDM) method (Becker et al., 1964). Intuitively, this procedure asks “What is the minimum amount you would need to be paid in order to agree to treatment?” in such a way that participants are incentivized to answer truthfully. The dollar-valued answer provides a continuous proxy of each participant’s propensity to self-select into treatment. Embedding this procedure into an experiment allows researchers to estimate how effects vary continuously across the selection distribution.

CHESS builds on a long-standing literature on self-selection in political science and medicine which proposes designs like the preference-incorporating choice and assignment (PICA) design and patient-preference trials (Torgerson and Sibbald, 1998; de Benedictis-Kessner et al., 2019). These designs ask participants to choose between treatment and a

control condition, allowing researchers to recover treatment effects for those who do and do not select into treatment. While natural, these designs have three key drawbacks. First, they treat self-selection propensity as discrete rather than continuous, yielding just two heterogeneous treatment effects for an experiment with one treatment and one control condition. Second, these designs are inefficient: they require large samples because participants must be divided between choice and random-assignment arms. Third, the conclusions of choice-based designs may shift sharply, sometimes even reversing, depending on the desirability of the researcher-supplied control condition. CHESS overcomes these limitations by eliciting a continuous measure of the propensity to self-select into treatment for each participant prior to randomization.

I illustrate CHESS with two preregistered online experiments studying pro- and anti-gun-control ads.¹ While average treatment effects suggest that both ads are equivalently effective at changing attitudes towards gun control, CHESS reveals that the pro-gun-control ad is most effective among participants who are likely to watch it, while the anti-gun-control ad is most effective among those who are not. Moreover, I find evidence of non-linear treatment effects on issue priority: the anti-gun-control ad reduces the priority of gun laws for participants with intermediate self-selection propensities. These results show how CHESS can yield practical insights that conventional designs would miss by enabling a more granular analysis of heterogeneity by self-selection propensity.²

Self-Selection in Political Communication

Political communication channels typically involve a degree of selective exposure. People may choose whether to browse BlueSky or Truth Social, whether to click on an article after seeing a headline, or whether to listen to a candidate speak on a podcast. Even in contexts where exposure is ostensibly unavoidable, such as with unskippable ads, people retain the choice of whether to pay attention or tune out.

Preferences for selective exposure to political information are well documented: people choose to consume news from ideologically congruent sources across many contexts (Taber and Lodge, 2006; Stroud, 2008; Kim, 2009; Iyengar and Hahn, 2009; Lavigne et al., 2025).

¹This experiment was reviewed and approved by the [REDACTED] IRB ID: [REDACTED]. Preregistrations are available at <https://aspredicted.org/fkmm-9qxz.pdf> and <https://aspredicted.org/wa6kq7.pdf>.

²AI Disclosure: Claude was used to assist in writing R code and for copyediting, and GPT-4 was used for classifying free responses.

It has long been theorized that preferences for selective exposure to news may shape voting decisions (Lazarsfeld et al., 1968). More recently, the rise of algorithmic social media feeds has raised concerns that preferences for selective exposure to ideologically congruent news may create ideological echo chambers contributing to political polarization (Iyengar et al., 2019; Ross Arguedas et al., 2022).

Despite the widespread acknowledgment that selective exposure mediates political communication, studies of the causal effect of communication interventions typically focus on average treatment effects using forced exposure designs (Feldman (2011); Levendusky (2013); Broockman and Kalla (2016); Kalla and Broockman (2022); Broockman and Kalla (2025), but see Kim and Patterson Jr (2025)).

Yet, estimating how persuasion varies with self-selection propensity is both practically and theoretically important. Practically, understanding self-selection is valuable for determining how effective an intervention will be. For example, when considering whether to run an ad campaign, it is useful to know both how likely people are to watch an ad and how persuasive that ad would be to those who watch it.

Theoretically, understanding self-selection can help us disentangle correlation from causation when answering core questions about political communication and persuasion. As one illustration, de Benedictis-Kessner et al. (2019) argue that estimating heterogeneity by self-selection is crucial for disentangling whether the observed tendency to self-select into congenial media is a cause of polarization or merely a reflection of it.

Existing Designs for Studying Self-Selection

A small but growing experimental design literature has developed methods for estimating heterogeneous treatment effects by the propensity to self-select into treatment.³ I call this set of experimental designs *choice designs*, because the main idea behind them is to incorporate participant choice over treatment and control directly into the experiment. In a typical choice design, participants are first randomized into forced-exposure and selective-exposure arms. Participants in the forced-exposure arm are then randomly assigned to treatment or control, as in a standard experiment. Participants in the selective-exposure arm choose

³The broader problem of recovering heterogeneous treatment effects by the propensity to self-select into treatment in quasi-experimental settings has been studied in the econometrics literature under the name marginal treatment effects (Björklund and Moffitt, 1987; Imbens and Angrist, 1994; Heckman and Vytlačil, 2005).

whether they would prefer to be assigned to treatment or control. Together, these arms allow researchers to estimate heterogeneous treatment effects for those who do and do not self-select into treatment.

The idea of incorporating choice via forced- and selective-exposure arms was first proposed by [Rücker \(1989\)](#), and has seen take-up in the medical literature, where it is referred to as a patient-preference trial ([Torgerson and Sibbald, 1998](#); [King et al., 2005](#)). A choice design was first applied to political communication by [Gaines and Kuklinski \(2011\)](#) who study attack ads. [Arceneaux et al. \(2012\)](#), [Arceneaux and Johnson \(2013\)](#), and [Levendusky \(2013\)](#) incorporate stated preferences into choice designs to study the effects of partisan media on attitudes toward the media, policy opinions and polarization, respectively. A more recent methodological improvement in this literature comes from [Knox et al. \(2019\)](#) and [de Benedictis-Kessner et al. \(2019\)](#) who formalize the incorporation of stated preferences into choice designs within a unified experimental framework called the Preference-Incorporating Choice and Assignment (PICA) design, an approach which enables sensitivity analysis with respect to measurement error.⁴ [Testa et al. \(2021\)](#) modify PICA to account for dynamic selection, and [Egami et al. \(2026\)](#) propose a placebo-augmented version of PICA to improve precision.

Directly incorporating participant choice into an experimental design is an intuitive and empirically successful way to estimate heterogeneity by self-selection propensity. However, choice designs have three important drawbacks. First, they treat self-selection propensity as discrete rather than continuous. For example, in a PICA design studying those who prefer to consume entertainment or news, participants are classified into two discrete bins: those who prefer entertainment and those who prefer news (e.g., [Kim and Patterson Jr \(2025\)](#)). However, in reality, this kind of self-selection preference is likely to be continuous: some people may strongly prefer to watch entertainment content (or news), some may mildly prefer it, and some may be indifferent. By inferring a preference for entertainment or news content via a single choice, these designs forgo collecting information on the intensity of the preference. Because choice designs treat self-selection propensity as discrete, they cannot detect nonlinearities in treatment effects. This is not a purely hypothetical problem: I find nonlinearities in the effects of gun control issue ads on the priority of gun control in [Section 7](#). In contrast, CHESS allows researchers to recover how treatment effects vary continuously

⁴In a concurrent working paper, [Levy et al. \(2025\)](#) deploys a similar BDM elicitation-based design to study persuasion in the context of the Israel-Gaza war. The two papers were developed independently and are complementary: this paper proposes a general methodological framework and connects it to previous choice-based designs, while [Levy et al. \(2025\)](#) provides additional evidence of the approach’s value in a distinct substantive domain.

with self-selection propensity.

Second, choice designs typically require large samples: [Egami et al. \(2026\)](#) notes that recent political science papers relying on PICA designs use $N > 7,000$ to achieve reasonable precision. One reason choice designs require large samples is because participants must be divided across choice and random-assignment arms. By measuring self-selection propensity before randomization, CHESS avoids this problem enabling more efficient estimation of heterogeneous treatment effects. In [Appendix A](#), I develop a potential outcomes framework that allows for a comparison of CHESS and PICA ([Neyman, 1923](#); [Rubin, 1974](#)). Using this framework, in [Appendix B](#), I show that CHESS achieves the same precision for K heterogeneous treatment effects with $\frac{K}{6}$ times as much data under some simplifying assumptions. So, if both methods are used to compute heterogeneous treatment effects for participants with high and low self-selection propensity, then CHESS requires a third as much data.⁵ In a recent methodological contribution, [Egami et al. \(2026\)](#) propose a placebo-augmented version of PICA (PICA-2) to more efficiently estimate some heterogeneous treatment effects. Even compared to the most efficiently estimated effects in PICA-2, CHESS offers efficiency gains: under some simplifying assumptions, I show in [Appendix B](#) that CHESS requires half as much data as PICA-2 to achieve the same level of precision when estimating binary heterogeneous treatment effects.

Third, the conclusions drawn from choice-based designs may be sensitive to the researcher-supplied control condition. For example, when testing the persuasiveness of a Fox News article, the researcher has a variety of potential control conditions to offer participants: they could choose a pure control task (e.g., do nothing for a fixed period of time), a moderate alternative (e.g., a CNN article), a democratic alternative (e.g., an MSNBC article), or even a fun alternative (e.g., an Onion article). While this may seem innocuous, it interacts in important ways with the choice design: by offering more attractive alternatives, the researcher can entice a larger share of participants to self-select out of reading the Fox News article. For some nonlinear treatment-effect-heterogeneity curves, the desirability of the control condition can change the sign and magnitude of the estimated heterogeneous treatment effects, thereby reversing the qualitative conclusions drawn from the choice experiment. I discuss this problem in more detail in [Appendix C](#). By estimating a measure of self-selection propensity that does not depend on the control condition, CHESS avoids this problem.

⁵To maintain the same degree of precision, the sample size of CHESS must linearly increase with the number of heterogeneous treatment effect bins. This relationship can be relaxed by making a functional form assumption regarding the shape of heterogeneous treatment effects (e.g. a second degree polynomial, as in [Figure 6](#)). In this case, all data can be applied to recover the parametric estimates, further improving efficiency.

The CHES Design

The CHES design proceeds in three steps. First, participants are asked “What is the minimum amount of money you would need to be paid in order for you to participate in treatment?” in such a way that they are incentivized to answer truthfully. Second, most participants take part in a standard experiment. Finally, heterogeneous treatment effects are estimated using the minimum payment as a continuous proxy for self-selection propensity.

The key insight behind CHES is that if each participant’s self-selection propensity were observable, then estimating heterogeneous treatment effects would be straightforward: after randomizing participants into treatment and control, we could estimate heterogeneous treatment effects just like we would for any other continuous observable (e.g., income, years of education).

The first step of CHES is to elicit a proxy for self-selection propensity. CHES reframes the question from “How likely would you be to select into treatment?” to the related question “What is the minimum amount you would need to be paid to participate in treatment?” The benefit of this reframing is that it allows the researcher to implement the question with real stakes. Just like with choice designs, making participants answer a real-stakes question mitigates biases that arise from answering hypothetical questions (Gaines and Kuklinski, 2011). However, unlike choice designs, this question can be asked to every participant (not just those in a selective-exposure arm), and the answer is a continuous measure of self-selection propensity rather than a discrete one.

Directly asking participants how much they would need to be paid could cause mis-measurement, since participants might strategically under- or over-report their answer.⁶ To overcome this problem, CHES elicits the minimum payment using the BDM procedure (Becker et al., 1964; Myerson, 1979). In this procedure, a participant’s reported minimum payment is compared to a randomly drawn wage. If the participant’s minimum payment is below the randomly drawn wage, then the participant is randomized into the experiment and paid the randomly drawn wage. The benefit of this procedure is that participants maximize their expected payoff by truthfully reporting the minimum amount that they would need to be paid. However, the cost is that the BDM mechanism is more difficult for participants to understand, so researchers should implement comprehension checks (Jack et al., 2022).⁷

⁶For example, participants may believe that overstating their minimum required payment would increase payment, or that understating it could increase the chance of selection.

⁷There are a few common willingness-to-pay elicitation procedures including standard BDM, multiple price lists, and binary search, each of which has costs and benefits. The stability and accuracy of willingness-

The second step of the CHES design is to conduct an experiment. However, in order to avoid deception, participants who say that they need to be paid more than the randomly drawn wage must be excluded from the experiment.⁸ This has some benefits: it prevents the coercion of research subjects, and may remove subjects who would not comply with treatment from the experiment. However, it means that CHES cannot estimate treatment effects for participants with a very high aversion to treatment. This limitation is shared with choice designs, where participants who strongly prefer to avoid treatment may attrit when assigned to the forced-exposure arm.

The third step of CHES is to compute heterogeneous treatment effects as normal. For example, researchers can discretize the minimum payment variable into K evenly sized bins to compute K heterogeneous treatment effects. Alternatively, the treatment effect can be estimated as a continuous function of the minimum payment.

Evaluating Gun-Control Issue Ads with CHES

Issue ads account for a meaningful fraction of political advertising dollars: [Falk et al. \(2006\)](#) find that \$404 million was spent on issue ads during the 108th congress. Yet, the prior literature on the effectiveness of television ads has largely focused on candidate ads ([Kalla and Broockman, 2022](#)). [Hall and Reynolds \(2012\)](#) theorize that issue ads may operate by increasing the salience of an issue or by changing public attitudes. [Kalla and Broockman \(2022\)](#) study ads regarding immigration and LGBTQ inclusion, and find that these ads are remembered but largely unpersuasive. [Junk and Rasmussen \(2024\)](#) study an advertising campaign about privacy risk, and similarly find null effects on attitudes and issue salience.

In this paper, I use CHES to study issue ads from two prominent campaigns related to gun control. Participants in the pro-gun-control treatment view [Back-To-School Essentials](#), an ad created by Sandy Hook Promise. Participants in the anti-gun-control treatment view [Freedom’s Safest Place — Real Empowerment](#), an ad created by the National Rifle Association. In a placebo condition, participants view a [PSA about texting while driving](#) produced by the news station WJBF.

to-pay elicitation mechanisms is a subject of ongoing research ([Jack et al., 2022](#)).

⁸The BDM procedure requires offering participants a price drawn from a random distribution. However, the vast majority of the weight of this distribution can be on the single price that the research team plans to pay participants. To avoid deception, a small number of participants can be offered another price, and ideally, the minimum and the maximum price described to participants should be offered.

I conducted two preregistered online survey experiments on Prolific in September 2025 and February 2026. In the first experiment, a total of 883 participants were recruited, restricted to U.S. residents aged 18-65.⁹ Participants who incorrectly answered comprehension questions regarding the BDM procedure twice were screened out of the survey. Participants were iteratively asked “Would you be willing to watch a 1 minute long advertisement for [bonus]?” using a binary search procedure to search over bonuses between 10 cents and 2 dollars following an implementation of BDM suggested by [Berkouwer and Dean \(2022\)](#). Participants were recruited into the experiment if they agreed to watch an ad for less than the randomly chosen bonus. In practice, nearly all participants were offered a \$1.00 bonus, which resulted in 93% of participants (823 out of 883) being recruited into the experiment, though 11 participants attrited before outcome collection.¹⁰ [Table 1](#) provides summary statistics by treatment arm for both experiments, and is consistent with successful randomization. Recruited participants were randomly shown one of the ads. To incentivize viewing the ad, payment was contingent on correctly answering two attention check questions related to the content of the ad. Finally, motivated by the theoretical channels proposed in [Hall and Reynolds \(2012\)](#), I asked participants to report their attitudes towards gun control as well as the priority of gun control relative to other issues.¹¹

The second experiment replicates the first, but includes three modifications designed to validate aspects of CHES: content-specific minimum bonus elicitation, a behavioral measure of ad avoidance, and a larger, more representative sample.

The first experiment only elicited a single measure of self-selection propensity under the assumption that there was a shared self-selection component across ads: people who tend to avoid ads will tend to avoid both pro- and anti-gun-control ads. In the second experiment, I elicited content-specific measures of self-selection propensity, which allowed for an explicit test of this assumption and enabled estimation of heterogeneity by content-specific self-selection propensity. I implemented the BDM mechanism by asking participants three slider

⁹I also place two additional restrictions on Prolific sample recruitment: I restrict to Prolific workers with a 99%+ approval rate on previous studies, who have not participated in any of this study’s pilots.

¹⁰The randomization procedure is a bonus of \$1.00 with 99% probability, and a bonus of \$2.00 with 1% probability. Randomization is used so that the claims that (1) the price is randomly chosen and that (2) the bonus may be up to \$2.00 are truthful.

¹¹Issue attitude is the participant’s support for gun control, as measured by a 7-point Likert scale question “In your opinion, should it be harder or easier to get a gun in this country?” with higher values indicating greater support for gun control. Issue priority is defined by the rank of “gun laws” in response to the question “Please rank the following issues from 1 (most important) to 5 (least important) when it comes to deciding your vote. The ranking should reflect how much the issue matters to you, no matter what position you personally take on it.” with the options (a) The Environment/Climate Change, (b) Immigration Policy, (c) Gun Laws, (d) Abortion Policy, and (e) The Economy. For figures and tables, I recode issue priority so that higher values indicate greater importance.

questions: “What is the minimum bonus you would need to be paid to watch a 1-minute (non-partisan ad/pro-gun-control ad/anti-gun-control ad)?” with a scale ranging from \$0.00 to \$1.26. To allow for randomization, participants were only recruited into the experiment if they were willing to watch all three ads for less than the randomly selected bonus. Nearly all participants were offered a \$1.10 bonus, and 81% of participants (1109 out of 1363) were recruited into the second experiment.¹²

The second modification was to collect a behavioral measure of ad-avoidance to test if the measure of self-selection propensity correlated with real behavior. Collecting a behavioral measure of when participants would like to skip an ad presents an experimental design challenge, because participants must watch the whole ad to measure causal treatment effects. I resolved this tension by telling participants “You are encouraged to skip the ad whenever you’d normally skip such an ad by clicking the yellow →, and you will still be paid your bonus. However, your skip may not work immediately as the computer has randomly selected a minimum watch time.” In practice, the randomly selected minimum watch time varies from 53 to 57 seconds, ensuring that participants are exposed to the ad, while capturing a behavioral measure of when they would have liked to skip the ad.

Because participants are told that they may skip some of the ad, I removed the requirement that participants must correctly answer two attention check questions in order to be paid a bonus. This likely resulted in less attention being paid to each ad, but made the experiment more externally valid. Because I anticipated smaller treatment effects, I preregistered a larger sample size. In addition to the sampling requirements from the first experiment, I recruited participants using quota sampling to select a population that is representative of the U.S. by gender and political affiliation.

CHESS Captures Variation in Selection Propensity

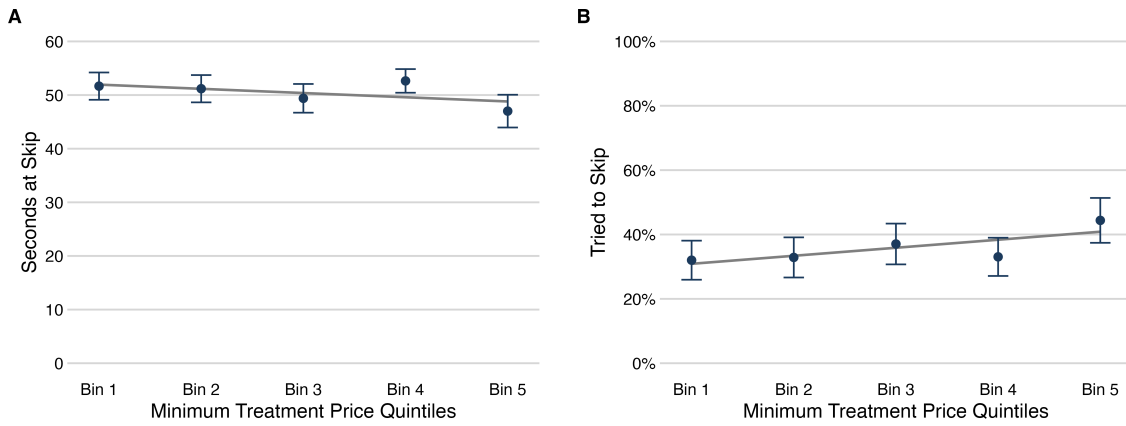
A key assumption of CHESS is that the reported minimum price captures meaningful variation in self-selection propensity.¹³ I use a behavioral measure of ad avoidance from the second experiment along with a variety of additional analyses to test that assumption. Overall, the results suggest that the minimum bonus is a useful proxy.

¹²The randomization procedure is \$1.10 with 99% probability, and \$1.26 with 1% probability.

¹³More broadly, validating how well laboratory measures of self-selection predict real-world behavior is an open question for both CHESS and choice designs, since both rely on lab elicitation to proxy for field behavior.

I preregister the hypothesis that the minimum bonus that must be paid to view an ad is negatively correlated with the amount of time spent watching that ad. Both mean price across the three treatment arms and the realized price from the arm that participants were randomly assigned to are negatively correlated with watch time (Spearman rank correlation average: -0.08 , $p=0.01$; realized: -0.05 , $p=0.07$). These are modest correlations, but they reflect meaningful variation in behavior: Figure 1 shows that those who report a top quintile minimum bonus watch about 10% less of the ad, and are about 40% more likely to attempt to skip the ad early.

Figure 1: Behavioral Measures of Selection Propensity by Minimum Treatment Price



Several additional checks show that the elicited minimum price is a useful proxy for self-selection propensity. First, participants self-reported preference for which ad they would prefer to watch matches their lowest minimum price 75% of the time. Second, participants self-reported propensity to skip ads is modestly positively correlated with minimum price ($r=0.06$, $p=0.01$). Third, more conservative participants require a higher price to watch the pro-gun control ad relative to the anti-gun control ad ($r = 0.236$, $p < 0.001$). Together, these correlations suggest that participants understand the BDM procedure and respond in sensible ways.

In order to validate the design choice of eliciting a single minimum price in the first experiment, I preregister the hypothesis that the minimum bonus elicitation will be positively correlated. I find strong evidence for this claim with intercorrelations ranging from $r=0.69$ to $r=0.81$ ($p<0.01$). This suggests that a large component of ad aversion is shared across ad content, so learning a participant's overall self-selection propensity is a useful proxy for content-specific self-selection propensity in the context of issue ads.

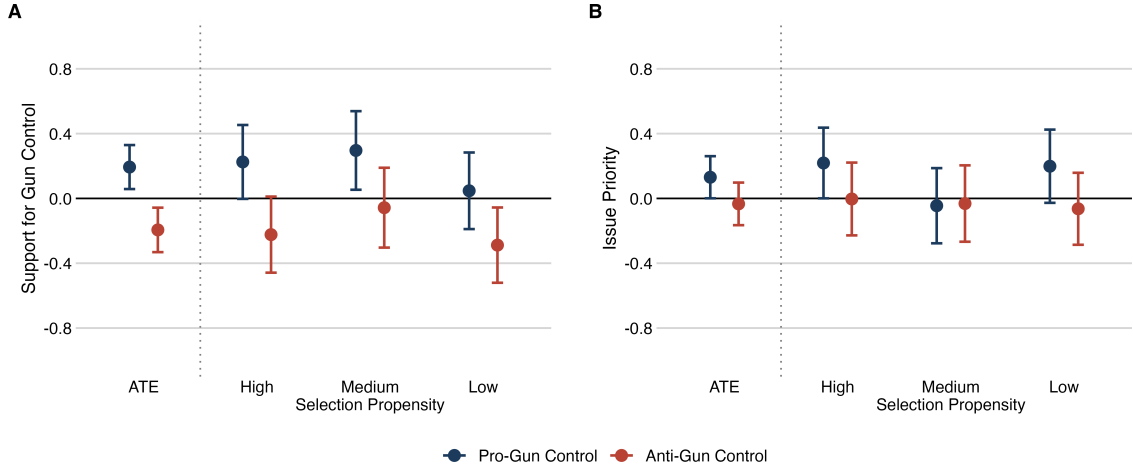
Experimental Results

The results showcase how average treatment effects can obscure practically important heterogeneity by self-selection propensity. In this section, I focus on results from a pooled sample and specifications with controls to maximize precision, but I report results with and without controls broken out by experiment in [Appendix D](#). I preregister hypotheses that the anti-gun-control ad will decrease support for gun control, the pro-gun-control ad will increase support for gun control, and that both ads will increase the issue priority of gun laws.¹⁴ [Figure 2](#) graphs average treatment effects, and [Table 2](#) reports average effects by experiment. Consistent with the first two hypotheses, the anti-gun-control ad reduces support by 0.19 points ($p < 0.01$) and the pro-gun-control ad increases support for gun control by 0.19 points ($p < 0.01$) on a 7 point Likert scale. These are meaningful treatment effects: they represent 15% of the baseline difference between liberal and conservative attitudes on gun control (liberal mean: 6.10 vs. conservative mean: 4.84), or about 13% of a standard deviation ($SD = 1.47$). I find no evidence that the anti-gun-control ad changes the priority of gun laws. However, the pro-gun-control ad increases issue priority by 0.13 ranks ($p = 0.05$) out of 5 issues, which is 11% of a standard deviation. So, if we were to evaluate these ads on the basis of average treatment effects, we could conclude that both ads are equivalently effective at changing attitudes towards gun control, and the pro-gun-control ad modestly increases issue priority.

However, heterogeneity by self-selection propensity tells a more nuanced story: the pro- and anti-gun control ads are most effective on opposite ends of the selection spectrum. [Figure 2](#) graphs heterogeneous treatment effects, and [Tables 3](#) and [4](#) report results by experiment. The pro-gun-control ad is particularly effective among the participants who are most willing to watch it. The ad increases support for gun control by 0.29 points ($p = 0.02$) for medium-propensity participants, and suggestively increases support by 0.23 points ($p = 0.052$) for high-propensity participants. It also increases the issue priority of gun laws by 0.22 ranks ($p = 0.05$) among high propensity participants, which is 70% larger than the average treatment effect. In contrast, the anti-gun-control ads effects are strongest on low-propensity participants: it reduces support by 0.29 points ($p = 0.02$), though it suggestively reduces support among high propensity participants as well (0.22 points, $p = 0.06$). The anti-gun-control

¹⁴Preregistrations are available at <https://aspredicted.org/fknm-9qxz.pdf> and <https://aspredicted.org/wa6kq7.pdf>. I do not cover all preregistered secondary analyses in this section due to space constraints, but I include every preregistered analysis in [Appendix D](#). In the second experiment, I hypothesize that only pro-gun-control ad will increase the issue priority of gun control based on the results of the first experiment.

Figure 2: Heterogeneous Treatment Effects by Self-Selection Propensity



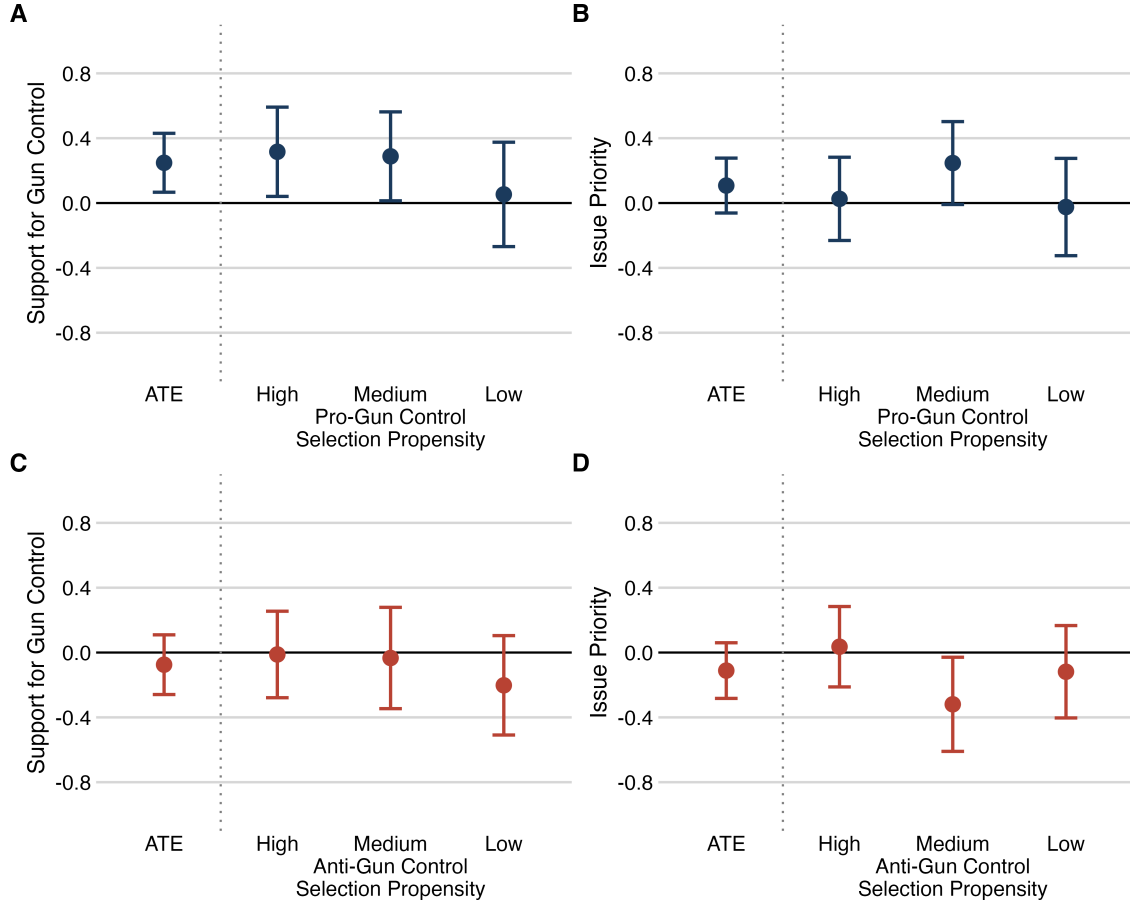
Notes: ATE is the average treatment effect for the whole sample. High, medium, and low represent terciles of the participant's elicited minimum bonus to view an advertisement, where high propensity corresponds to the lowest payment. Points are regression coefficients, and error bars are 95% confidence intervals.

ad has no effect on issue priority across the selection spectrum.

Measures of content-specific self-selection propensity reinforce the idea that the pro- and anti-gun-control ads change attitudes towards gun control on opposite ends of the selection spectrum. [Figure 3](#), [Table 5](#) and [Table 6](#) analyze data from Experiment 2, where participants provide a measure of content-specific self-selection propensity by indicating the minimum bonus they would need to watch pro-gun-control, anti-gun-control, and non-partisan ads. The pro-gun-control ad increases support for gun control by 0.35 points ($p=0.03$) among the participants most willing to watch pro-gun-control ads. In contrast, the anti-gun-control ad appears most effective among those least willing to watch anti-gun-control ads, though the effect is imprecisely estimated (-0.28 points, $p=0.10$). These effects sizes are meaningful: they represent roughly a quarter of the baseline difference between liberal and conservative attitudes towards gun control. Taken together, these analyses show how CHESS can lead researchers to make qualitatively different inferences: while average treatment effects suggest that the ads have similar effects on attitudes, heterogeneity by selection propensity reveals that they have different effects on the people who are most likely to view them, leading to different conclusions about efficacy in the field and suggesting different optimal targeting strategies.

Interestingly, the content-specific self-selection propensity results also show that the anti-gun control ad reduces the issue priority of gun laws by 0.34 ranks ($p=0.04$) among partici-

Figure 3: Heterogeneous Treatment Effects by Content-Specific Selection Propensity



Notes: Experiment 2 sample. High, medium, and low are terciles of participant's minimum bonus to view a (pro-/anti-gun-control) ad, where higher propensity corresponds to lower payments. Points are regression coefficients, and error bars are 95% confidence intervals.

pants with a medium propensity to view anti-gun-control ads, contrary to the hypothesized direction of the effect. This result is both methodologically and theoretically interesting. Methodologically, nonlinear treatment effects highlight the value of CHES, because they cannot be recovered by choice designs which only allow for binary comparisons (e.g., high vs. low propensity). Theoretically, the issue priority channel proposed in [Hall and Reynolds \(2012\)](#)'s account of outside lobbying suggests that ads may raise the priority of issues. However, this result could still be consistent with the spirit of the outside lobbying theory if anti-gun-control lobbyists wanted to lower the priority of gun control, because they were satisfied with lax status quo gun laws.

Even though the ads each have clear ideological valences, self-selection propensity is not just a proxy for political ideology. An analysis of heterogeneity by ideology in [Table 7](#) shows

that the anti-gun-control ad is persuasive to conservatives, while the pro-gun-control ad is persuasive to moderates. Estimating treatment effects separately within liberal, moderate and conservative subsamples in [Table 8](#) shows that the anti-gun-control ad is most effective among low-propensity conservatives, while the pro-gun-control ad is most effective among high-propensity moderates. These results show that heterogeneity by self-selection propensity can uncover insights that heterogeneity by political ideology alone would miss.

Substantively, the results suggest that issue ads can both change attitudes and change issue priorities, at least for some subsets of the population, providing empirical support for the mechanisms described in [Hall and Reynolds \(2012\)](#)’s theory of outside lobbying. While these results are intuitive, they do contrast with findings from the small prior experimental literature on issue ads which tends to find null effects ([Kalla and Broockman, 2022](#); [Junk and Rasmussen, 2024](#)). However, I study different ads about different issues, measure outcomes online rather than in the field, and find effects that vary with selection propensity, so many channels could potentially drive these differences.

Conclusion

I propose a new experimental design, Continuous Heterogeneity Estimation for Self-Selection (CHESS). The design connects an old literature on willingness-to-pay elicitation with more recent work on leveraging participant choice to uncover heterogeneity by self-selection propensity ([Becker et al., 1964](#); [de Benedictis-Kessner et al., 2019](#)).

CHESS offers three advantages over existing designs. First, while choice designs restrict researchers to estimating binary heterogeneous treatment effects, CHESS treats self-selection propensity as continuous. This enables estimation of rich patterns of heterogeneity including non-linear treatment effect curves that cannot be uncovered by choice designs. Second, CHESS elicits a proxy for self-selection propensity for every participant prior to randomization. This makes CHESS more efficient, requiring a third of the data to produce comparable estimates to choice designs. Given the large samples typically required for choice designs, this makes CHESS more accessible to researchers operating with limited resources. Third, CHESS reduces the sensitivity of conclusions to the desirability of the control condition, an issue that can reverse the qualitative conclusions of choice designs. However, these advantages come at the cost of increased experimental complexity: CHESS requires the inclusion of additional comprehension checks, and is less straightforward to administer.

I study issue ads using the CHES design, and find support for [Hall and Reynolds \(2012\)](#)'s theory of outside lobbying: issue ads change both attitudes and issue priority, at least for some participants. CHES uncovers patterns of heterogeneity that change conclusions about the efficacy of these ads. While average treatment effects imply that the pro-gun-control ad and the anti-gun-control ad are equally effective, CHES reveals that the pro-gun-control ad is effective among respondents who are likely to watch it, while the anti-gun-control ad is effective among those who are not.

Overall, CHES can provide researchers with more granular insights into treatment effect heterogeneity in contexts like political communication where participant self-selection is an important concern. By allowing researchers to estimate effects on participants who are likely to opt into treatment, CHES improves the external validity of experimental findings.

I conclude with two directions for future research. First, validating how well laboratory measures of self-selection predict real-world behavior remains an open empirical question. Both choice designs and CHES assume that choices in the lab predict choices in the field, but direct evidence for this assumption is limited.

Second, there are many plausible ways to refine the CHES design. For example, money may not be the most externally valid metric for measuring self-selection propensity: the BDM procedure could be extended to capture non-monetary costs such as time (e.g., "How long would you wait to avoid watching an ad?"). More broadly, the BDM procedure could be replaced with other behavioral proxies for self-selection propensity. Because lab experiments inform some of the highest stakes political communications decisions, it would be valuable to make progress on improving the external validity of these experimental designs.

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9 Tables

Table 1: Balance Table

	Experiment 1			Experiment 2		
	Control	Anti	Pro	Control	Anti	Pro
<i>Politics</i>						
Ideology	3.519 (0.104)	3.436 (0.115)	3.374 (0.107)	3.525 (0.091)	3.653 (0.092)	3.802 (0.089)
Political Engagement	4.220 (0.139)	4.220 (0.146)	4.252 (0.149)	4.391 (0.118)	4.379 (0.120)	4.225 (0.116)
<i>Demographics</i>						
Female	0.590 (0.029)	0.510 (0.031)	0.551 (0.030)	0.512 (0.026)	0.466 (0.027)	0.439 (0.026)
Black	0.113 (0.019)	0.106 (0.019)	0.150 (0.022)	0.110 (0.016)	0.102 (0.016)	0.083 (0.014)
Hispanic	0.078 (0.016)	0.080 (0.017)	0.086 (0.017)	0.063 (0.012)	0.071 (0.014)	0.080 (0.014)
Asian	0.126 (0.019)	0.106 (0.019)	0.120 (0.020)	0.084 (0.014)	0.045 (0.011)	0.080 (0.014)
Native	0.007 (0.005)	0.034 (0.011)	0.015 (0.007)	0.029 (0.009)	0.028 (0.009)	0.027 (0.008)
<i>Education</i>						
High School	0.307 (0.027)	0.308 (0.029)	0.345 (0.029)	0.354 (0.025)	0.311 (0.025)	0.350 (0.025)
College	0.509 (0.029)	0.490 (0.031)	0.453 (0.031)	0.486 (0.026)	0.525 (0.027)	0.489 (0.026)
Post-Graduate	0.184 (0.023)	0.202 (0.025)	0.202 (0.025)	0.160 (0.019)	0.164 (0.020)	0.160 (0.019)
Observations	293	263	267	381	354	374

Notes: Entries are sample means of covariates by treatment arm, with standard errors in parentheses. Ideology ranges from 1 (very liberal) to 7 (very conservative). Political engagement ranges from 0 (Never) to 7 (Every day of the week).

Table 2: Average Treatment Effects

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Controls:	No	Yes	No	Yes
<i>Experiment 1</i>				
Constant	5.680*	6.493*	2.952*	2.837*
	(0.0844)	(0.1716)	(0.0715)	(0.1713)
Anti-Gun Control	-0.3715*	-0.3554*	0.0597	0.0790
	(0.1231)	(0.1048)	(0.1042)	(0.1046)
Pro-Gun Control	0.1631	0.1150	0.1512	0.1489
	(0.1227)	(0.1042)	(0.1039)	(0.1041)
Observations	812	812	812	812
<i>Experiment 2</i>				
Constant	5.320*	6.153*	2.982*	3.130*
	(0.0746)	(0.1529)	(0.0607)	(0.1425)
Anti-Gun Control	-0.1422	-0.0749	-0.1200	-0.1114
	(0.1075)	(0.0940)	(0.0875)	(0.0876)
Pro-Gun Control	0.1129	0.2484*	0.1066	0.1078
	(0.1060)	(0.0927)	(0.0863)	(0.0864)
Observations	1,109	1,109	1,109	1,109
<i>Pooled</i>				
Constant	5.476*	6.285*	2.969*	3.002*
	(0.0562)	(0.1143)	(0.0463)	(0.1095)
Anti-Gun Control	-0.2429*	-0.1943*	-0.0438	-0.0336
	(0.0814)	(0.0700)	(0.0670)	(0.0671)
Pro-Gun Control	0.1260	0.1937*	0.1256	0.1308*
	(0.0806)	(0.0693)	(0.0664)	(0.0663)
Observations	1,921	1,921	1,921	1,921

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, political ideology, and degree of political engagement. Attitude is support for gun control on a 7-point scale where higher means more supportive. Issue Priority is the rank of gun laws among five policy issues, rescaled so higher values mean greater priority.

Table 3: Heterogeneous Treatment Effects on Attitude by Self-Selection Propensity

	Experiment 1		Experiment 2		Pooled	
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	5.596*	6.415*	5.362*	6.235*	5.470*	6.296*
	(0.1384)	(0.1918)	(0.1290)	(0.1791)	(0.0948)	(0.1309)
Anti-Gun Control × High Propensity	-0.3882	-0.4307*	-0.2081	-0.0469	-0.3003*	-0.2234
	(0.2055)	(0.1749)	(0.1885)	(0.1650)	(0.1393)	(0.1198)
Anti-Gun Control × Medium Propensity	-0.3892	-0.3536	0.1594	0.1116	-0.0281	-0.0571
	(0.2408)	(0.2042)	(0.1833)	(0.1602)	(0.1463)	(0.1257)
Anti-Gun Control × Low Propensity	-0.3386	-0.2831	-0.4049*	-0.3066	-0.3769*	-0.2882*
	(0.2016)	(0.1707)	(0.1863)	(0.1638)	(0.1374)	(0.1185)
Pro-Gun Control × High Propensity	0.1829	0.1858	0.1021	0.2745	0.1217	0.2253
	(0.1987)	(0.1681)	(0.1832)	(0.1612)	(0.1350)	(0.1163)
Pro-Gun Control × Medium Propensity	0.1253	0.1277	0.2638	0.3608*	0.2558	0.2961*
	(0.2305)	(0.1950)	(0.1846)	(0.1616)	(0.1440)	(0.1238)
Pro-Gun Control × Low Propensity	0.1434	-0.0474	-0.0662	0.0891	-0.0107	0.0472
	(0.2185)	(0.1850)	(0.1832)	(0.1606)	(0.1403)	(0.1206)
High Propensity	0.1514	0.1892	0.1208	-0.0558	0.1410	0.0637
	(0.1948)	(0.1650)	(0.1859)	(0.1628)	(0.1350)	(0.1161)
Medium Propensity	0.1079	0.1385	-0.2225	-0.2065	-0.1370	-0.0908
	(0.2203)	(0.1861)	(0.1794)	(0.1570)	(0.1386)	(0.1192)
Controls	No	Yes	No	Yes	No	Yes
R ²	0.02477	0.31567	0.01432	0.25726	0.01621	0.27978
Observations	812	812	1,109	1,109	1,921	1,921

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, political ideology, and degree of political engagement. Attitude is support for gun control on a 7-point scale where higher means more supportive. Selection propensity is a tercile split of the minimum bonus to watch a one minute ad, where higher propensity corresponds to a lower bonus. Terciles are computed within-experiment for pooling. Placebo × Low Propensity is the omitted category.

Table 4: Heterogeneous Treatment Effects on Issue Priority by Self-Selection Propensity

	Experiment 1		Experiment 2		Pooled	
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	3.028*** (0.1168)	2.942*** (0.1915)	2.906*** (0.1049)	3.044*** (0.1663)	2.962*** (0.0780)	2.993*** (0.1252)
Anti-Gun Control × High Propensity	0.0495 (0.1736)	0.0615 (0.1747)	-0.0677 (0.1533)	-0.0504 (0.1533)	-0.0127 (0.1146)	-0.0034 (0.1146)
Anti-Gun Control × Medium Propensity	0.1561 (0.2033)	0.2009 (0.2039)	-0.1306 (0.1491)	-0.1553 (0.1489)	-0.0280 (0.1204)	-0.0315 (0.1203)
Anti-Gun Control × Low Propensity	0.0137 (0.1703)	0.0190 (0.1705)	-0.1705 (0.1515)	-0.1344 (0.1522)	-0.0880 (0.1131)	-0.0639 (0.1134)
Pro-Gun Control × High Propensity	0.3340** (0.1677)	0.3333** (0.1678)	0.1036 (0.1490)	0.1055 (0.1497)	0.2071* (0.1111)	0.2187** (0.1113)
Pro-Gun Control × Medium Propensity	0.1290 (0.1946)	0.1528 (0.1947)	-0.1344 (0.1502)	-0.1485 (0.1501)	-0.0366 (0.1185)	-0.0449 (0.1184)
Pro-Gun Control × Low Propensity	-0.0001 (0.1845)	-0.0338 (0.1847)	0.3185** (0.1490)	0.3316** (0.1492)	0.1897 (0.1155)	0.1988* (0.1153)
High Propensity	-0.0546 (0.1645)	-0.0712 (0.1648)	0.1538 (0.1512)	0.1651 (0.1513)	0.0556 (0.1111)	0.0628 (0.1111)
Medium Propensity	-0.2247 (0.1860)	-0.2391 (0.1858)	0.0798 (0.1459)	0.0887 (0.1458)	-0.0392 (0.1141)	-0.0275 (0.1140)
Controls	No	Yes	No	Yes	No	Yes
R ²	0.01070	0.02909	0.01548	0.03241	0.00946	0.02048
Observations	812	812	1,109	1,109	1,921	1,921

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, political ideology, and degree of political engagement. Issue Priority is the rank of gun laws among five policy issues, rescaled so higher values mean greater priority. Selection propensity is a tercile split of the minimum bonus to watch a one minute ad, where higher propensity corresponds to a lower bonus. Terciles are computed within-experiment for pooling. Placebo × Low Propensity is the omitted category.

Table 5: Heterogeneous Treatment Effects by Pro-Gun Control Selection Propensity

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.050*	6.041*	2.980*	3.092*
	(0.1444)	(0.1925)	(0.1184)	(0.1795)
Pro-Gun Control $\times$ High Propensity	0.2748	0.3477*	0.0178	0.0125
	(0.1765)	(0.1551)	(0.1447)	(0.1446)
Pro-Gun Control $\times$ Medium Propensity	0.0280	0.2525	0.2242	0.2201
	(0.1701)	(0.1500)	(0.1395)	(0.1398)
Pro-Gun Control $\times$ Low Propensity	-0.0500	0.0697	0.0200	0.0337
	(0.2063)	(0.1818)	(0.1692)	(0.1694)
High Propensity	0.4460*	0.1548	0.1480	0.1814
	(0.1937)	(0.1715)	(0.1588)	(0.1599)
Medium Propensity	0.3026	0.1077	-0.1146	-0.0757
	(0.1850)	(0.1641)	(0.1517)	(0.1530)
Controls	No	Yes	No	Yes
R ²	0.02760	0.26032	0.01315	0.02986
Observations	1,109	1,109	1,109	1,109

Notes: Experiment 2 Sample. OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Indicators for Anti-Gun Control $\times$ High/Medium/Low Propensity are included in all specifications but suppressed for clarity. Controls: race, gender, educational attainment, political ideology, and degree of political engagement. Selection propensity is a tercile split of the minimum bonus to watch a one minute *pro-gun-control* ad, where higher propensity corresponds to a lower minimum bonus.

Table 6: Heterogeneous Treatment Effects by Anti-Gun Control Selection Propensity

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.513*	6.303*	2.904*	3.041*
	(0.1360)	(0.1816)	(0.1103)	(0.1687)
Anti-Gun Control $\times$ High Propensity	-0.0643	0.0246	-0.0115	-0.0117
	(0.1690)	(0.1477)	(0.1371)	(0.1372)
Anti-Gun Control $\times$ Medium Propensity	-0.0544	-0.00002	-0.3511*	-0.3392*
	(0.1993)	(0.1737)	(0.1617)	(0.1614)
Anti-Gun Control $\times$ Low Propensity	-0.3355	-0.2846	-0.0539	-0.0358
	(0.1959)	(0.1713)	(0.1589)	(0.1592)
High Propensity	-0.2433	-0.2186	0.1483	0.1580
	(0.1802)	(0.1574)	(0.1462)	(0.1462)
Medium Propensity	-0.3201	-0.2267	0.0606	0.0726
	(0.1927)	(0.1690)	(0.1564)	(0.1570)
Controls	No	Yes	No	Yes
R ²	0.00830	0.25602	0.01437	0.03045
Observations	1,109	1,109	1,109	1,109

Notes: Experiment 2 Sample. OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Indicators for Pro-Gun Control $\times$ High/Medium/Low Propensity are included in all specifications but suppressed for clarity. Controls: race, gender, educational attainment, political ideology, and degree of political engagement. Selection propensity is a tercile split of the minimum bonus to watch a one minute *anti-gun-control* ad, where higher propensity corresponds to a lower minimum bonus.

Table 7: Heterogeneous Treatment Effects by Political Ideology

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.069*	4.577*	3.034*	3.035*
	(0.1068)	(0.1290)	(0.0997)	(0.1226)
Anti-Gun Control $\times$ Liberal	-0.1868	-0.1543	-0.0416	-0.0393
	(0.1033)	(0.1014)	(0.0965)	(0.0964)
Anti-Gun Control $\times$ Moderate	0.0781	0.0950	-0.1448	-0.1377
	(0.1535)	(0.1508)	(0.1434)	(0.1434)
Anti-Gun Control $\times$ Conservative	-0.6138*	-0.5656*	0.0247	0.0491
	(0.1317)	(0.1295)	(0.1230)	(0.1231)
Pro-Gun Control $\times$ Liberal	0.0979	0.1156	0.1569	0.1533
	(0.1040)	(0.1021)	(0.0972)	(0.0971)
Pro-Gun Control $\times$ Moderate	0.4278*	0.4704*	0.0243	0.0389
	(0.1490)	(0.1465)	(0.1392)	(0.1393)
Pro-Gun Control $\times$ Conservative	0.0154	0.0422	0.1548	0.1637
	(0.1291)	(0.1269)	(0.1206)	(0.1206)
Liberal	1.064*	1.006*	-0.0500	-0.0353
	(0.1285)	(0.1264)	(0.1201)	(0.1201)
Conservative	-0.3435*	-0.3261*	-0.1374	-0.1222
	(0.1396)	(0.1371)	(0.1305)	(0.1303)
Controls	No	Yes	No	Yes
R ²	0.23290	0.26662	0.00467	0.01509
Observations	1,921	1,921	1,921	1,921

Notes: Pooled Sample. OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, and degree of political engagement.

Table 8: HTEs on Attitude by Political Ideology $\times$ Selection Propensity

	Attitude					
	Liberal		Moderate		Conservative	
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	6.197*	5.620*	4.902*	4.284*	4.679*	4.375*
	(0.1041)	(0.1463)	(0.2173)	(0.2660)	(0.1612)	(0.2192)
Anti-Gun Control $\times$ High Propensity	-0.3270*	-0.2916	-0.2089	-0.0289	-0.5874*	-0.5096
	(0.1519)	(0.1496)	(0.2771)	(0.2713)	(0.2673)	(0.2618)
Anti-Gun Control $\times$ Medium Propensity	0.0686	0.0453	0.1448	0.1327	-0.3012	-0.3559
	(0.1635)	(0.1610)	(0.3057)	(0.2981)	(0.2560)	(0.2529)
Anti-Gun Control $\times$ Low Propensity	-0.2604	-0.2276	0.3617	0.2883	-0.9034*	-0.7913*
	(0.1559)	(0.1536)	(0.2894)	(0.2868)	(0.2371)	(0.2351)
Pro-Gun Control $\times$ High Propensity	0.1107	0.1322	0.4546	0.6199*	-0.0625	-0.0719
	(0.1512)	(0.1496)	(0.2734)	(0.2676)	(0.2447)	(0.2415)
Pro-Gun Control $\times$ Medium Propensity	0.3490*	0.3400*	0.4031	0.4885	0.1803	0.1662
	(0.1654)	(0.1635)	(0.2881)	(0.2834)	(0.2507)	(0.2474)
Pro-Gun Control $\times$ Low Propensity	-0.1625	-0.1462	0.5537	0.5440	-0.1701	-0.0974
	(0.1589)	(0.1565)	(0.2849)	(0.2813)	(0.2534)	(0.2488)
High Propensity	-0.0377	-0.0459	0.3762	0.2138	0.3041	0.3267
	(0.1506)	(0.1483)	(0.2810)	(0.2769)	(0.2433)	(0.2397)
Medium Propensity	-0.1663	-0.1381	0.0278	-0.0085	-0.1410	-0.0639
	(0.1537)	(0.1516)	(0.3037)	(0.2974)	(0.2391)	(0.2357)
Controls	No	Yes	No	Yes	No	Yes
R ²	0.01977	0.06570	0.02984	0.10585	0.05917	0.11913
Observations	911	911	434	434	576	576

Notes: Pooled Sample. OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, and political engagement.

A Potential Outcomes Statistical Framework Connecting PICA and CHESS

In order to more precisely compare the CHESS and PICA designs, I now formalize the estimation problem using a potential outcomes framework (Neyman, 1923; Rubin, 1974). Let $X_i \in \{0, 1\}$ indicate whether individual i receives treatment, let Y_i denote individual i 's observed outcome, and let $C_i \in \{0, 1\}$ denote whether individual i chooses to participate in the intervention if given a choice. Let $Y_i(1)$ and $Y_i(0)$ denote individual i 's potential outcomes under treatment and control, respectively.

In a standard PICA design, participants are randomly assigned to one of two arms:

1. A *choice arm*, where participants self-select into the intervention (C_i is observed and determines X_i);
2. A *forced-assignment arm*, where participants are assigned to treatment or control regardless of preference (X_i is randomly assigned).

This design allows us to identify the following treatment effects:

1. $E[Y_i(1) - Y_i(0)]$, the average treatment effect
2. $E[Y_i(1) - Y_i(0) \mid C_i = 1]$, the treatment effect among participants who would opt into treatment if given a choice
3. $E[Y_i(1) - Y_i(0) \mid C_i = 0]$, the treatment effect among participants who would opt into control if given a choice

Let z denote a participant's willingness-to-select-into-treatment, which we can think of as their anticipated utility from participation.

In the choice arm, the researcher offers participants the choice to either opt into treatment or to select an outside option. For now, assume that all participants value the outside option at a fixed level, d . Then,

$$C_i = \mathbb{I}[z_i > d]$$

That is, participants opt into treatment if they like treatment more than the outside option. Implicitly, the provided alternative divides participants into high- and low-willingness-to-select-into-treatment groups. Then, the PICA heterogeneous treatment effects correspond

to

$$\begin{aligned} E[Y_i(1) - Y_i(0) \mid z > d] \\ E[Y_i(1) - Y_i(0) \mid z \leq d], \end{aligned}$$

where the value of d depends on how desirable the researcher-supplied outside option is to participants.

Suppose that the elicitation procedure provides a valid measure of participants' willingness-to-select-into-treatment. That is, assume that the answer to "How much would I need to pay you to participate in treatment?" perfectly correlates with willingness-to-select-into-treatment z_i . Then, we can compute

$$E[Y_i(1) - Y_i(0) \mid z \in (a, b)]$$

for any interval (a, b) with empirical support in the data. In particular, we can recover the PICA estimates, as well as any other intervals of interest.

In practice, we may want to provide a non-parametric estimate for how the treatment effect varies with willingness-to-select-into-treatment. To do this, we can define cutpoints along z that divide the data into K bins, and then can compute K within-bin treatment effects.

B Precision of CHESS/PICA/PICA-2 Estimators

This section derives back-of-the-envelope standard errors for (i) a CHESS estimator that reports K heterogeneous treatment effects (HTEs) and (ii) a binary-treatment PICA estimator that reports two choice-specific HTEs (those who would select into treatment vs. those who would not). I'll show that, under some simplifying assumptions, CHESS requires $\frac{K}{6}$ times as many observations to achieve the same degree of precision as PICA for each HTE.

B.1 Standard Error of Treatment Effect Estimator

First, I will derive the standard error of a treatment effect estimator. Let $X \in \{0, 1\}$ indicate the treatment condition, and let Y be the outcome variable. Suppose we have N observations, randomized evenly into treatment and control. The treatment effect estimate is given by

$$\tau = \mathbb{E}[Y|X = 1] - \mathbb{E}[Y|X = 0]$$

with estimator

$$\hat{\tau} = \frac{2}{N} \sum_{i=1}^N y_i \mathbb{I}[X = 1] - \frac{2}{N} \sum_{i=1}^N y_i \mathbb{I}[X = 0].$$

Then,

$$\text{Var}(\hat{\tau}) = \text{Var}(\bar{Y}_1 - \bar{Y}_0) = \text{Var}(\bar{Y}_1) + \text{Var}(\bar{Y}_0).$$

Define

$$\sigma_0^2 = \text{Var}(Y|X = 0) \quad \sigma_1^2 = \text{Var}(Y|X = 1).$$

Then,

$$\text{Var}(\bar{Y}_1) = \frac{2\sigma_1^2}{N} \quad \text{Var}(\bar{Y}_0) = \frac{2\sigma_0^2}{N}$$

Assuming homoskedasticity of errors,

$$\sigma^2 = \sigma_0^2 = \sigma_1^2$$

so

$$\text{Var}(\hat{\tau}) = \text{Var}(\bar{Y}_1) + \text{Var}(\bar{Y}_0) = \frac{2\sigma_1^2}{N} + \frac{2\sigma_0^2}{N} = \frac{4\sigma^2}{N}$$

and

$$\text{SE}(\hat{\tau}) = 2\sigma\sqrt{\frac{1}{N}}.$$

B.2 Efficiency of CHESSE Estimator

Now, I will derive the standard error for heterogeneous treatment effects when we divide N observations into K equal sized bins, representing K equal cuts along some dimension of heterogeneity Z . In each bin, there are $\frac{N}{K}$ observations. As a simplifying assumption, I impose homoskedasticity of errors within each of these bins. Formally, $\forall k \in K$,

$$\sigma^2 = \text{Var}(Y \mid X = 0, Z \in \text{bin } k) = \text{Var}(Y \mid X = 1, Z \in \text{bin } k)$$

Then,

$$\text{Var}(\hat{\tau}_k) = \frac{4K\sigma^2}{N},$$

and

$$\text{SE}(\hat{\tau}_k) = 2\sigma\sqrt{\frac{K}{N}}.$$

So, if a researcher divides the willingness-to-select spectrum into K equally-sized bins and collects N_{CHESSE} observations, then

$$\text{SE}(\hat{\tau}_{\text{CHESSE}}) = 2\sigma\sqrt{\frac{K}{N_{\text{CHESSE}}}}.$$

B.3 Efficiency of PICA Estimator

Following [Knox et al. \(2019\)](#), let $C \in \{0, 1\}$ indicate the treatment condition that a participant would choose if given the option. Then, the estimator for the PICA heterogeneous treatment effects in the binary case is given by

$$\tau_1 = \mathbb{E}[Y(1) - Y(0) \mid C = 1] = \frac{\mu_{\text{sel}} - \mu_{\text{forced},0}}{p},$$

and

$$\tau_0 = \mathbb{E}[Y(1) - Y(0) \mid C = 0] = \frac{\mu_{\text{forced},1} - \mu_{\text{sel}}}{1 - p},$$

where p represents the probability that participants select into treatment in the selective-exposure arm, and C represents the choice that participants make in the selective-exposure

arm. This estimator can be found after equation four in [Knox et al. \(2019\)](#) and matches the estimator in [Gaines and Kuklinski \(2011\)](#).

Let total PICA sample size be N_{PICA} . Assume equal allocation across arms:

$$\text{forced arm size} = \frac{N_{\text{PICA}}}{2}, \quad \text{selective arm size} = \frac{N_{\text{PICA}}}{2}.$$

Within the forced arm, assume treatment is randomized 50/50, so

$$n_{1,\text{forced}} = \frac{N_{\text{PICA}}}{4}, \quad n_{0,\text{forced}} = \frac{N_{\text{PICA}}}{4}.$$

To get PICA estimators, replace the population means with sample means:

$$\hat{\tau}_1 = \frac{\hat{\mu}_{\text{sel}} - \hat{\mu}_{\text{forced},0}}{p}, \quad \hat{\tau}_0 = \frac{\hat{\mu}_{\text{forced},1} - \hat{\mu}_{\text{sel}}}{1-p}.$$

with variances

$$\text{Var}(\hat{\tau}_1) = \text{Var}\left[\frac{\hat{\mu}_{\text{sel}} - \hat{\mu}_{\text{forced},0}}{p}\right] = \frac{1}{p^2}(\text{Var}[\hat{\mu}_{\text{sel}}] + \text{Var}[\hat{\mu}_{\text{forced},0}])$$

and

$$\text{Var}(\hat{\tau}_0) = \text{Var}\left[\frac{\hat{\mu}_{\text{forced},1} - \hat{\mu}_{\text{sel}}}{1-p}\right] = \frac{1}{(1-p)^2}(\text{Var}[\hat{\mu}_{\text{forced},1}] + \text{Var}[\hat{\mu}_{\text{sel}}])$$

where I use the fact that the means are estimated on disjoint sets of observations to ignore covariance.

Under the assumptions above along with homoskedasticity of errors, we know that for a sample of size N , $\text{Var}(\sum_{i=1}^N y_i) = \frac{\sigma^2}{N}$. So, we can estimate the variances of each of the terms in the estimator:

$$\begin{aligned} \text{Var}(\hat{\mu}_{\text{sel}}) &= \frac{\sigma^2}{N_{\text{PICA}}/2} = \frac{2\sigma^2}{N_{\text{PICA}}}, \\ \text{Var}(\hat{\mu}_{\text{forced},0}) &= \frac{\sigma^2}{N_{\text{PICA}}/4} = \frac{4\sigma^2}{N_{\text{PICA}}}, \end{aligned}$$

and

$$\text{Var}(\hat{\mu}_{\text{forced},1}) = \frac{4\sigma^2}{N_{\text{PICA}}}.$$

Treating p as fixed,

$$\text{Var}(\hat{\tau}_1) = \frac{1}{p^2} (\text{Var}(\hat{\mu}_{\text{sel}}) + \text{Var}(\hat{\mu}_{\text{forced},0})) = \frac{1}{p^2} \left(\frac{2\sigma^2}{N_{\text{PICA}}} + \frac{4\sigma^2}{N_{\text{PICA}}} \right) = \frac{6\sigma^2}{p^2 N_{\text{PICA}}},$$

so

$$\text{SE}(\hat{\tau}_1) = \frac{\sqrt{6} \sigma}{p \sqrt{N_{\text{PICA}}}}.$$

Similarly,

$$\text{Var}(\hat{\tau}_0) = \frac{1}{(1-p)^2} (\text{Var}(\hat{\mu}_{\text{forced},1}) + \text{Var}(\hat{\mu}_{\text{sel}})) = \frac{6\sigma^2}{(1-p)^2 N_{\text{PICA}}},$$

so

$$\text{SE}(\hat{\tau}_0) = \frac{\sqrt{6} \sigma}{(1-p) \sqrt{N_{\text{PICA}}}}.$$

To obtain a symmetric benchmark where the two PICA HTEs have equal precision, assume $p = 0.5$. Then,

$$\text{SE}(\hat{\tau}_{\text{PICA}}) = \text{SE}(\hat{\tau}_1) = \text{SE}(\hat{\tau}_0) = \frac{\sqrt{6} \sigma}{(1/2) \sqrt{N_{\text{PICA}}}} = 2\sigma \sqrt{\frac{6}{N_{\text{PICA}}}}.$$

B.4 Comparing Efficiency of PICA and CHESS Estimators

Suppose we have N_{PICA} observations for a PICA design and we select K groups for CHESS. I solve for the number of observations N_{CHESS}^* required to estimate treatment effects with equal precision to PICA:

$$\begin{aligned} \text{SE}(\hat{\tau}_{\text{PICA}}) &= \text{SE}(\hat{\tau}_{\text{CHESS}}) \\ 2\sigma \sqrt{\frac{6}{N_{\text{PICA}}}} &= 2\sigma \sqrt{\frac{K}{N_{\text{CHESS}}}} \\ \frac{N_{\text{CHESS}}}{N_{\text{PICA}}} &= \frac{K}{6} \\ N_{\text{CHESS}}^* &= \frac{K}{6} N_{\text{PICA}} \end{aligned}$$

So, we need $\frac{K}{6}$ times as many observations in order for each CHESS heterogeneous treatment effect to achieve the same level of precision as each of the two PICA heterogeneous treatment effects.

Note that this is a back-of-the-envelope discussion of estimator precision. In particular, I am ignoring any additional variance created by measuring a proxy for self-selection rather

observing the true self-selection propensity. I am also ignoring the loss of sample in CHESS due to participants having high minimum payment requirements.

B.5 Comparison to PICA-2 Estimator

I now compare the efficiency of CHESS to the PICA-2 estimator of [Egami et al. \(2026\)](#) under similar simplifying assumptions. I model a simplified version of their design with a single treatment (Chinese propaganda) and a control condition. Let total sample size be $N_{\text{PICA-2}}$, split evenly between a forced exposure arm ($D = 1$) and a free choice arm ($D = 0$), each of size $N_{\text{PICA-2}}/2$.

Forced exposure arm. Participants are randomly assigned to treatment or control with equal probability $1/2$, yielding:

$$n_{\text{forced},0} = N_{\text{PICA-2}}/4$$

$$n_{\text{forced},1} = N_{\text{PICA-2}}/4$$

Free choice arm. Participants choose $C \in \{0, 1\}$ with probabilities $1 - p$ and p respectively. Those who choose $C = 1$ are further randomized 50/50 into treatment or placebo. Those who choose $C = 0$ receive the control condition directly. For this back-of-the-envelope calculation, I assume $p = 1/2$. This yields:

$$n_{\text{free},C=0} = N_{\text{PICA-2}}/4$$

$$n_{\text{free},C=1,T=1} = N_{\text{PICA-2}}/8$$

$$n_{\text{free},C=1,T=1_P} = N_{\text{PICA-2}}/8$$

Chooser ACTE. Following [Egami et al. \(2026\)](#) Appendix A, the chooser ACTE is identified as:

$$E(Y(1) - Y(0)|C = 1) = E(Y|T = 1, C = 1, D = 0) - E(Y|T = 1_P, C = 1, D = 0)$$

The plug-in estimator is:

$$\hat{\tau}_1^{\text{PICA-2}} = \bar{Y}_{\text{free},C=1,T=1} - \bar{Y}_{\text{free},C=1,T=1_P}$$

Each component is estimated on a disjoint sample of size $N_{\text{PICA-2}}/8$, so under homoskedasticity:

$$\begin{aligned}\text{Var}(\bar{Y}_{\text{free}, C=1, T=1}) &= \frac{\sigma^2}{N_{\text{PICA-2}}/8} = \frac{8\sigma^2}{N_{\text{PICA-2}}} \\ \text{Var}(\bar{Y}_{\text{free}, C=1, T=1_P}) &= \frac{\sigma^2}{N_{\text{PICA-2}}/8} = \frac{8\sigma^2}{N_{\text{PICA-2}}}\end{aligned}$$

Since the two groups are disjoint:

$$\begin{aligned}\text{Var}(\hat{\tau}_1^{\text{PICA-2}}) &= \frac{8\sigma^2}{N_{\text{PICA-2}}} + \frac{8\sigma^2}{N_{\text{PICA-2}}} = \frac{16\sigma^2}{N_{\text{PICA-2}}} \\ \text{SE}(\hat{\tau}_1^{\text{PICA-2}}) &= \frac{4\sigma}{\sqrt{N_{\text{PICA-2}}}} = 2\sigma\sqrt{\frac{4}{N_{\text{PICA-2}}}}\end{aligned}$$

Comparing to the PICA chooser SE at $p = 0.5$:

$$\frac{\text{SE}(\hat{\tau}_1^{\text{PICA-2}})}{\text{SE}(\hat{\tau}_1^{\text{PICA}})} = \frac{2\sigma\sqrt{4/N_{\text{PICA-2}}}}{2\sigma\sqrt{6/N_{\text{PICA}}}} = \sqrt{\frac{4}{6}} \approx 0.816$$

So, PICA-2 produces standard errors that are about 18% shorter than PICA in this simpler framework.

Non-chooser ACTE. Following [Egami et al. \(2026\)](#) Appendix A, the non-chooser ACTE is given by:

$$E(Y(1) - Y(0)|C = 0) = \frac{E(Y(1) - Y(0)) - E(Y(1) - Y(0)|C = 1) \cdot p}{1 - p}$$

which is estimated by the plug-in estimator:

$$\hat{\tau}_0^{\text{PICA-2}} = \frac{\hat{\tau}_{\text{ATE}} - \hat{\tau}_1^{\text{PICA-2}} \cdot p}{1 - p}$$

where $\hat{\tau}_{\text{ATE}} = \bar{Y}_{\text{forced}, T=1} - \bar{Y}_{\text{forced}, T=0}$ is estimated from the $N_{\text{PICA-2}}/4$ forced-arm participants assigned to $T = 1$ and the $N_{\text{PICA-2}}/4$ assigned to $T = 0$, and $\hat{\tau}_1^{\text{PICA-2}}$ is the chooser ACTE estimator derived above. Treating p as fixed and using the fact that the forced and free

choice arms are disjoint:

$$\text{Var}(\hat{\tau}_{\text{ATE}}) = \frac{\sigma^2}{N_{\text{PICA-2}}/4} + \frac{\sigma^2}{N_{\text{PICA-2}}/4} = \frac{8\sigma^2}{N_{\text{PICA-2}}}$$

At $p = 1/2$:

$$\begin{aligned} \text{Var}(\hat{\tau}_0^{\text{PICA-2}}) &= \frac{1}{(1-p)^2} (\text{Var}(\hat{\tau}_{\text{ATE}}) + p^2 \cdot \text{Var}(\hat{\tau}_1^{\text{PICA-2}})) \\ &= \frac{1}{(1/2)^2} \left(\frac{8\sigma^2}{N_{\text{PICA-2}}} + (1/2)^2 \cdot \frac{16\sigma^2}{N_{\text{PICA-2}}} \right) \\ &= 4 \left(\frac{8\sigma^2}{N_{\text{PICA-2}}} + \frac{4\sigma^2}{N_{\text{PICA-2}}} \right) = \frac{48\sigma^2}{N_{\text{PICA-2}}} \\ \text{SE}(\hat{\tau}_0^{\text{PICA-2}}) &= 2\sigma \sqrt{\frac{12}{N_{\text{PICA-2}}}} \end{aligned}$$

Comparing to the PICA non-chooser SE at $p = 0.5$, setting $N_{\text{PICA-2}} = N_{\text{PICA}} = N$:

$$\frac{\text{SE}(\hat{\tau}_0^{\text{PICA-2}})}{\text{SE}(\hat{\tau}_0^{\text{PICA}})} = \frac{2\sigma \sqrt{12/N}}{2\sigma \sqrt{6/N}} = \sqrt{\frac{12}{6}} = \sqrt{2} \approx 1.41$$

So the PICA-2 non-chooser ACTE standard error is $\sqrt{2}$ times as long as the PICA non-chooser ACTE standard error — approximately 41% longer.

Comparison to CHESS. I now compare the precision of CHESS to PICA-2. For the chooser ACTE, setting $\text{SE}(\hat{\tau}_1^{\text{PICA-2}}) = \text{SE}(\hat{\tau}_{\text{CHESS}})$:

$$\begin{aligned} 2\sigma \sqrt{\frac{4}{N_{\text{PICA-2}}}} &= 2\sigma \sqrt{\frac{K}{N_{\text{CHESS}}}} \\ \frac{N_{\text{CHESS}}}{N_{\text{PICA-2}}} &= \frac{K}{4} \\ N_{\text{CHESS}}^* &= \frac{K}{4} N_{\text{PICA-2}} \end{aligned}$$

So CHESS requires $\frac{K}{4}$ times as many observations as PICA-2 to achieve the same precision for the chooser ACTE. For the non-chooser ACTE, setting $\text{SE}(\hat{\tau}_0^{\text{PICA-2}}) = \text{SE}(\hat{\tau}_{\text{CHESS}})$:

$$\begin{aligned} 2\sigma\sqrt{\frac{12}{N_{\text{PICA-2}}}} &= 2\sigma\sqrt{\frac{K}{N_{\text{CHESS}}}} \\ \frac{N_{\text{CHESS}}}{N_{\text{PICA-2}}} &= \frac{K}{12} \\ N_{\text{CHESS}}^* &= \frac{K}{12}N_{\text{PICA-2}} \end{aligned}$$

So CHESS requires only $\frac{K}{12}$ times as many observations as PICA-2 to achieve the same precision for the non-chooser ACTE.

Taken together, relative to PICA-2, CHESS requires between $\frac{K}{12}$ and $\frac{K}{4}$ times as many observations depending on which ACTE is the quantity of interest. For $K = 2$ bins, CHESS requires between $\frac{1}{6}$ and $\frac{1}{2}$ times as many observations as PICA-2.

Note that this comparison abstracts away from several important differences between the designs. I do not account for participants whose high willingness-to-pay excludes them from CHESS. Moreover, CHESS does not require a valid placebo assumption, and enables continuously heterogeneity estimation rather than discrete estimation.

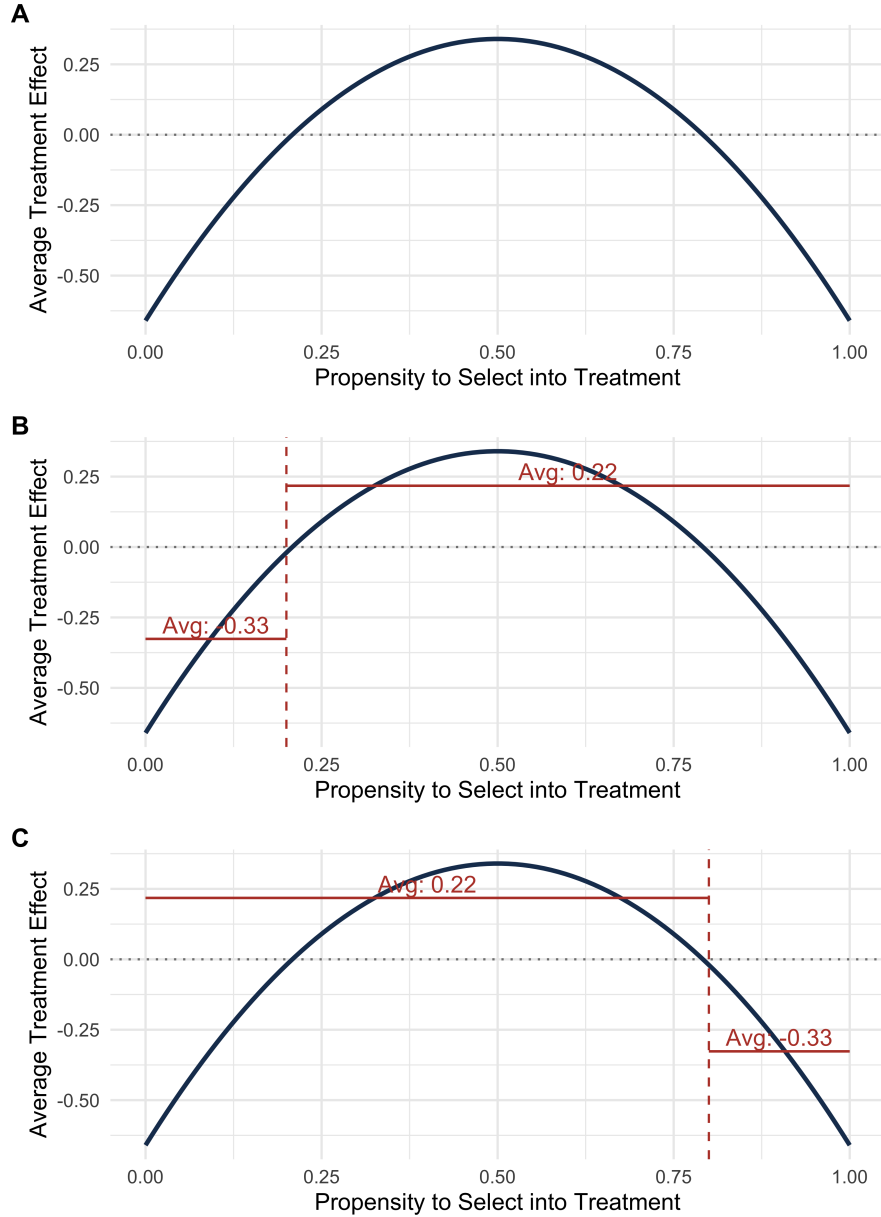
C Appendix: Reversal of PICA Heterogeneous Treatment Effects

In this section, I argue that the heterogeneous treatment effects estimated by PICA and related choice designs may be sensitive to the choice of control condition in such a way that it could reverse the qualitative conclusions drawn from the analysis. In choice designs, researchers estimate heterogeneous treatment effects for groups of participants who, in the selective-exposure condition, opt into and out of treatment. One obvious lever of control that researchers have is the desirability of the control condition. For example, in an experiment measuring the persuasiveness of a Fox News article, the control condition could be unpleasant (e.g., do nothing for a long, fixed period of time), or it could be fun (e.g., read an Entertainment article). Even ignoring that different control conditions could induce different groups of participants to opt out of treatment, it is clear that researchers can manipulate the share of participants who opt out of treatment. This degree of control is problematic because, for some non-linear heterogeneous treatment effect curves, the qualitative conclusions that researchers draw can reverse, since both the sign and magnitude of the estimated heterogeneous treatment effect curves are sensitive to the fraction of participants who opt out of treatment.

[Figure 4](#) illustrates this problem with a hypothetical example. In the first panel, I graph a hypothetical heterogeneous treatment effect curve which is non-linear and largest for participants with a moderate self-selection propensity. The second and third panels of [Figure 4](#) simulate the heterogeneous treatment effects estimated by PICA when the researcher-supplied control is relatively more or less desirable. In panel two, the control condition is more desirable, so those with moderate self-selection propensity opt out, leading the researcher to conclude that the treatment is less effective for those who are likely to select into treatment. In panel three, the control condition is less desirable, so those with moderate self-selection propensity opt in, leading to the opposite conclusion. In this example, changing the desirability of the control condition reverses the conclusion of the PICA analysis.

While the hypothetical example described in [Figure 4](#) is constructed to be a worst-case scenario for choice-based designs, such non-linear treatment effects may arise in natural political communication contexts. For example, consider a piece of right-leaning content. Suppose Republicans are most willing to watch it, moderates somewhat willing, and Democrats unwilling. Additionally, suppose that the content is ineffective for Republicans (who already hold these views) and Democrats (who distrust the source), but persuasive for moderates (who are open to changing their views and do not reflexively distrust the source). This set

Figure 4: Hypothetical Heterogeneous Treatment Effects Curve



Notes: Panel A plots a hypothetical treatment effect curve as a function of propensity to self-select into treatment. Panels B and C illustrate how varying the outside option in a choice design produces different average treatment effect estimates across groups. Red dashed lines indicate the cutoff, and red horizontal lines represent average treatment effects on each side of the cutoff.

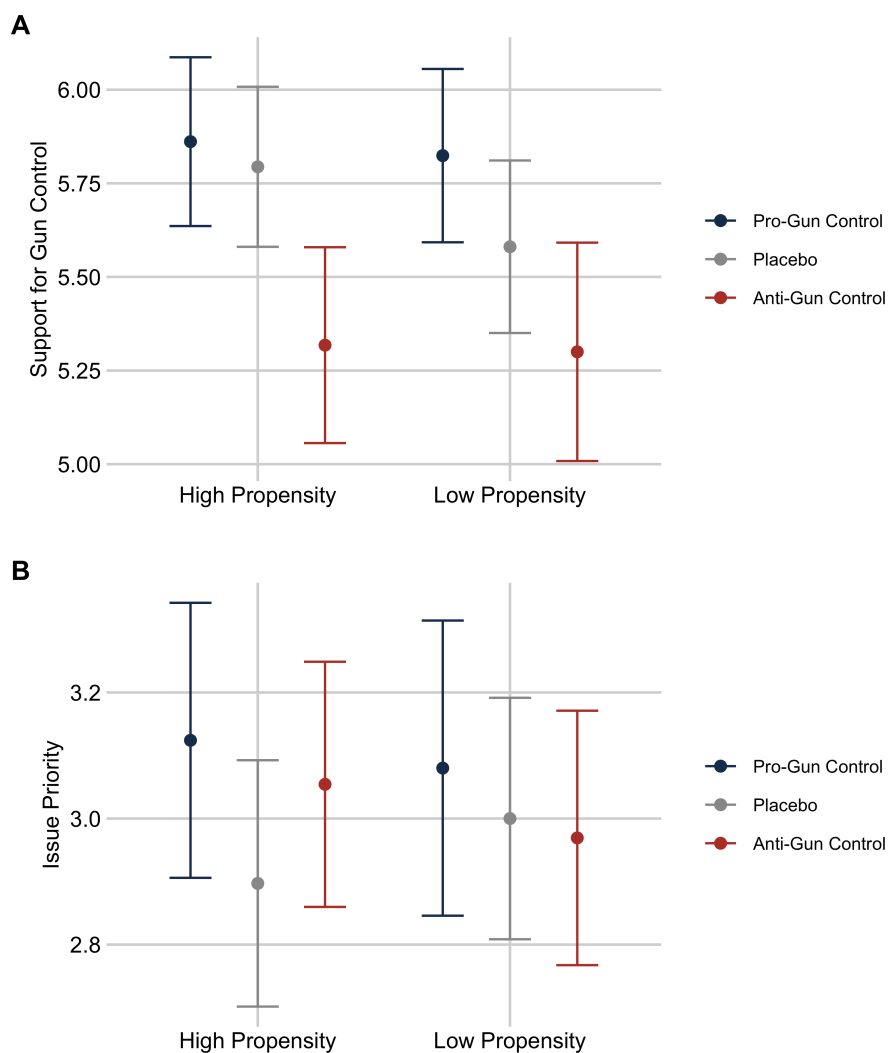
of circumstances generates exactly the kind of non-linear treatment effects curve described in [Figure 4](#).

Sensitivity to the control condition is mechanically avoided by CHES, because CHES

estimates self-selection propensity into treatment for each participant directly, rather than via choice between treatment and control. Moreover, by eliciting a continuous proxy for self-selection propensity, CHESS enables the estimation and detection of non-linear heterogeneous treatment effect curves directly.

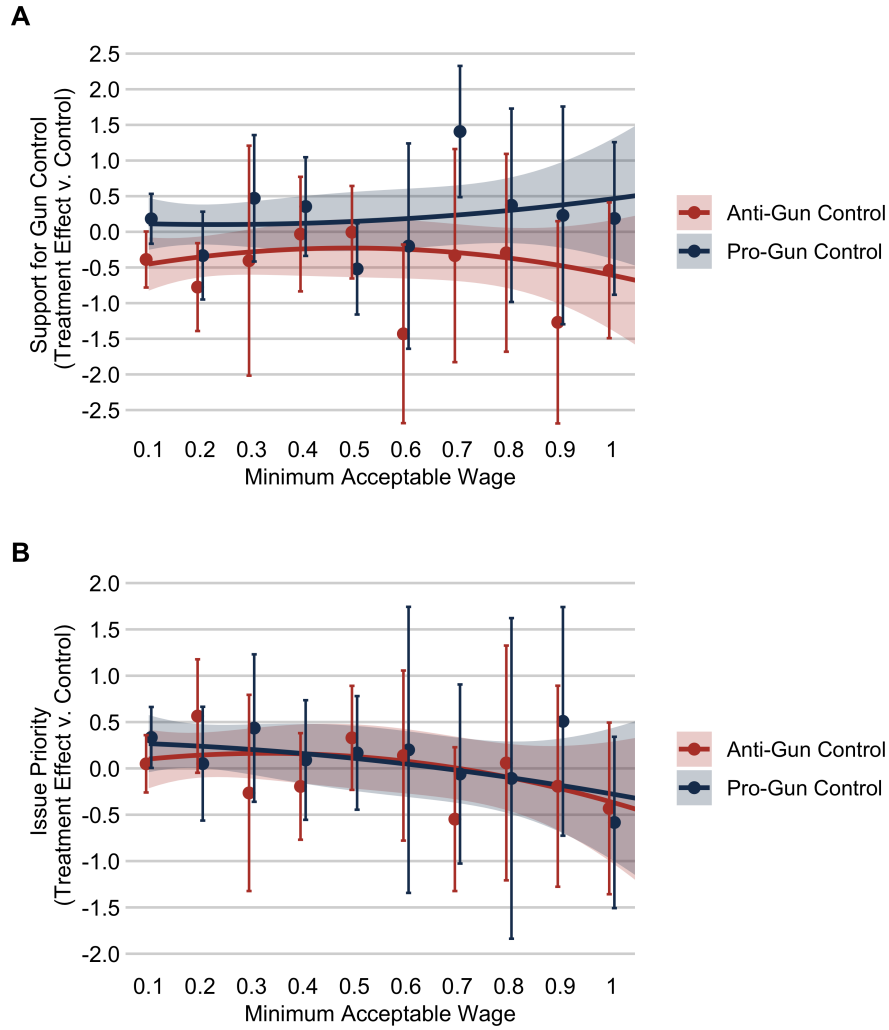
D Additional Results

Figure 5: Heterogeneity by Self-Selection into Treatment (Binary)



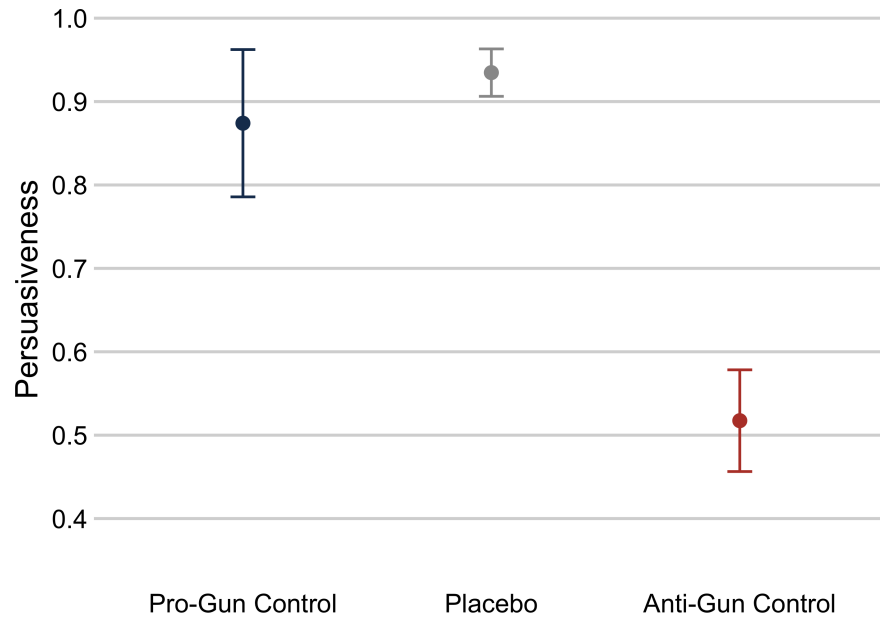
Notes: Experiment 1 sample. This figure replicates [Figure 2](#) with selection propensity divided into 2 bins rather than 3, making this approach more comparable to choice designs. Points represent means, and error bars represent 95% confidence intervals.

Figure 6: Heterogeneity by Self-Selection into Treatment (Parametric)



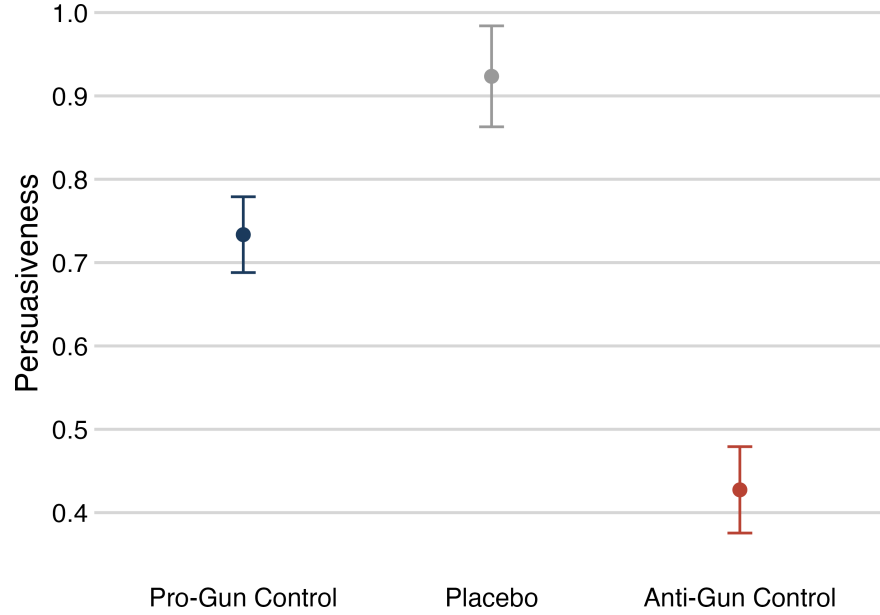
Notes: Experiment 1 sample. This figure plots treatment effects as a function of propensity to self-select into treatment. Curves are estimated using a second-degree polynomial interaction between treatment assignment and BDM elicited minimum compensation. Points represent bin-level treatment effects, and error bars represent 95% confidence intervals.

Figure 7: Perceived Persuasiveness of Ads (Experiment 1)



Notes: This figure graphs participant-reported persuasiveness of the ad by treatment arm. Persuasiveness is coded from participant free responses to the question “Did you find the ad persuasive? Why or why not?” using GPT-4 as a classifier, where 1 represents persuasive and 0 represents not persuasive. I validate coding by hand-coding a random subset of 100 responses. I agree with GPT-4’s coding in 95 out of 100 cases. However, an additional 4 cases are ambiguous, where GPT-4 and I agree but a reasonable coder might not. Points represent means, and error bars represent 95% confidence intervals.

Figure 8: Perceived Persuasiveness of Ads (Experiment 2)



Notes: This figure graphs participant-reported persuasiveness of the ad by treatment arm. Persuasiveness is coded from participant free responses to the question “Did you find the ad persuasive? Why or why not?” using GPT-4 as a classifier, where 1 represents persuasive and 0 represents not persuasive. Points represent means, and error bars represent 95% confidence intervals.

Table 9: Heterogeneous Treatment Effects by Political Ideology (Experiment 1)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.213*	4.725*	2.852*	2.771*
	(0.1582)	(0.1933)	(0.1565)	(0.1930)
Anti-Gun Control $\times$ Liberal	-0.2569	-0.2440	0.0787	0.0857
	(0.1476)	(0.1461)	(0.1459)	(0.1459)
Anti-Gun Control $\times$ Moderate	0.1387	0.1909	0.1105	0.1166
	(0.2309)	(0.2297)	(0.2283)	(0.2295)
Anti-Gun Control $\times$ Conservative	-1.083*	-0.9800*	-0.0143	0.0357
	(0.2000)	(0.1990)	(0.1978)	(0.1988)
Pro-Gun Control $\times$ Liberal	0.1608	0.1283	0.1782	0.1574
	(0.1484)	(0.1472)	(0.1468)	(0.1470)
Pro-Gun Control $\times$ Moderate	0.3323	0.3602	0.3748	0.3595
	(0.2195)	(0.2181)	(0.2170)	(0.2179)
Pro-Gun Control $\times$ Conservative	-0.0655	0.0059	-0.1094	-0.0612
	(0.2051)	(0.2036)	(0.2028)	(0.2034)
Liberal	4.3088*	1.036*	0.1133	0.1078
	(0.1884)	(0.1865)	(0.1863)	(0.1863)
Conservative	-0.2726	-0.3003	0.1475	0.1394
	(0.2051)	(0.2036)	(0.2028)	(0.2034)

Table 10: Heterogeneous Treatment Effects by Political Ideology (Experiment 2)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	4.964*	4.495*	3.167*	3.243*
	(0.1430)	(0.1730)	(0.1293)	(0.1592)
Anti-Gun Control $\times$ Liberal	-0.1293	-0.0869	-0.1411	-0.1552
	(0.1422)	(0.1397)	(0.1286)	(0.1285)
Anti-Gun Control $\times$ Moderate	0.0479	0.0284	-0.3252	-0.3087
	(0.2034)	(0.2003)	(0.1839)	(0.1843)
Anti-Gun Control $\times$ Conservative	-0.2998	-0.2778	0.0557	0.0865
	(0.1734)	(0.1708)	(0.1568)	(0.1571)
Pro-Gun Control $\times$ Liberal	0.0436	0.0976	0.1392	0.1220
	(0.1434)	(0.1411)	(0.1297)	(0.1298)
Pro-Gun Control $\times$ Moderate	0.4955*	0.5594*	-0.2356	-0.2009
	(0.2004)	(0.1976)	(0.1813)	(0.1818)
Pro-Gun Control $\times$ Conservative	0.0994	0.0817	0.3068*	0.2984*
	(0.1662)	(0.1633)	(0.1503)	(0.1502)
Liberal	1.030*	0.9782*	-0.1667	-0.1298
	(0.1736)	(0.1715)	(0.1570)	(0.1578)
Conservative	-0.3893*	-0.3446	-0.3417*	-0.3220
	(0.1864)	(0.1833)	(0.1686)	(0.1686)
Controls	No	Yes	No	Yes
R ²	0.19918	0.23817	0.01159	0.02687
Observations	1,109	1,109	1,109	1,109

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, and degree of political engagement.

Table 11: Heterogeneous Treatment Effects by Political Engagement (Experiment 1)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.685*	6.691*	2.863*	2.728*
	(0.1682)	(0.1896)	(0.1431)	(0.1892)
Anti-Gun Control $\times$ 0–3 Days	-0.5913*	-0.4501*	0.0466	0.0651
	(0.1879)	(0.1603)	(0.1599)	(0.1600)
Anti-Gun Control $\times$ 4–6 Days	-0.2732	-0.3460	0.1664	0.2038
	(0.2422)	(0.2071)	(0.2061)	(0.2067)
Anti-Gun Control $\times$ 7 Days	-0.1557	-0.2361	-0.0080	-0.0033
	(0.2185)	(0.1871)	(0.1858)	(0.1867)
Pro-Gun Control $\times$ 0–3 Days	0.0675	0.1230	0.1957	0.1771
	(0.1888)	(0.1611)	(0.1606)	(0.1609)
Pro-Gun Control $\times$ 4–6 Days	0.0062	-0.0234	0.1223	0.1432
	(0.2422)	(0.2061)	(0.2061)	(0.2058)
Pro-Gun Control $\times$ 7 Days	0.4057	0.2156	0.1235	0.1206
	(0.2150)	(0.1837)	(0.1829)	(0.1834)
0–3 Days	-0.0209	-0.0807	0.1450	0.1093
	(0.2117)	(0.1809)	(0.1801)	(0.1805)
7 Days	0.0140	0.0230	0.0832	0.1093
	(0.2248)	(0.1915)	(0.1912)	(0.1911)
Controls	No	Yes	No	Yes
R ²	0.03455	0.31075	0.00575	0.02271
Observations	812	812	812	812

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, and political ideology.

Table 12: Heterogeneous Treatment Effects by Political Engagement (Experiment 2)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.523*	6.305*	2.865*	2.875*
	(0.1374)	(0.1614)	(0.1123)	(0.1507)
Anti-Gun Control $\times$ 0–3 Days	0.0597	0.0492	-0.1221	-0.1258
	(0.1732)	(0.1517)	(0.1416)	(0.1416)
Anti-Gun Control $\times$ 4–6 Days	-0.2793	-0.1092	-0.0180	0.0106
	(0.1944)	(0.1708)	(0.1588)	(0.1594)
Anti-Gun Control $\times$ 7 Days	-0.2592	-0.2012	-0.2152	-0.2069
	(0.1901)	(0.1665)	(0.1554)	(0.1554)
Pro-Gun Control $\times$ 0–3 Days	0.5000*	0.5110*	-0.0383	-0.0552
	(0.1678)	(0.1469)	(0.1371)	(0.1371)
Pro-Gun Control $\times$ 4–6 Days	-0.3461	-0.0737	0.1687	0.1918
	(0.1911)	(0.1683)	(0.1561)	(0.1571)
Pro-Gun Control $\times$ 7 Days	0.1149	0.2424	0.2327	0.2546
	(0.1931)	(0.1690)	(0.1578)	(0.1577)
0–3 Days	-0.5225*	-0.2584	0.2721	0.2837
	(0.1824)	(0.1607)	(0.1490)	(0.1500)
7 Days	-0.0064	0.0159	0.0384	0.0279
	(0.1892)	(0.1660)	(0.1546)	(0.1549)
Controls	No	Yes	No	Yes
R ²	0.02193	0.26029	0.01421	0.02722
Observations	1,109	1,109	1,109	1,109

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, and political ideology.

Table 13: Heterogeneous Treatment Effects by Race (Experiment 1)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.966*	6.667*	2.816*	2.747*
	(0.1820)	(0.2113)	(0.1529)	(0.2086)
Anti-Gun Control $\times$ White	-0.2571	-0.2656*	0.0460	0.0658
	(0.1445)	(0.1235)	(0.1214)	(0.1220)
Anti-Gun Control $\times$ Black	-0.7459*	-0.9144*	0.2015	0.1679
	(0.3694)	(0.3154)	(0.3104)	(0.3113)
Anti-Gun Control $\times$ Other	-0.4877	-0.3847	0.2056	0.2242
	(0.2792)	(0.2386)	(0.2347)	(0.2356)
Pro-Gun Control $\times$ White	0.1983	0.1869	-0.0956	-0.0778
	(0.1459)	(0.1247)	(0.1226)	(0.1232)
Pro-Gun Control $\times$ Black	0.2149	-0.0261	0.8359*	0.7980*
	(0.3384)	(0.2893)	(0.2844)	(0.2856)
Pro-Gun Control $\times$ Other	-0.0964	-0.0286	0.5534*	0.5592*
	(0.2792)	(0.2378)	(0.2347)	(0.2347)
White	-0.4220*	-0.2144	0.1484	0.1343
	(0.2053)	(0.1762)	(0.1725)	(0.1739)
Black	0.1027	0.2461	0.1401	0.1264
	(0.3001)	(0.2568)	(0.2522)	(0.2536)
Controls	No	Yes	No	Yes
R ²	0.03584	0.30573	0.03126	0.03723
Observations	812	812	812	812

Notes: Pooled Sample. OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: gender, educational attainment, political ideology, and degree of political engagement.

Table 14: Heterogeneous Treatment Effects by Race (Experiment 2)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.744*	6.554*	2.972*	3.160*
	(0.2049)	(0.2273)	(0.1670)	(0.2120)
Anti-Gun Control $\times$ White	-0.0614	-0.0249	-0.1830	-0.1795
	(0.1187)	(0.1038)	(0.0968)	(0.0968)
Anti-Gun Control $\times$ Black	-0.0893	-0.0407	-0.1058	-0.0558
	(0.3320)	(0.2905)	(0.2707)	(0.2709)
Anti-Gun Control $\times$ Other	-0.3442	-0.3338	0.2617	0.2489
	(0.3354)	(0.2935)	(0.2735)	(0.2737)
Pro-Gun Control $\times$ White	0.1331	0.2335*	0.0732	0.0736
	(0.1180)	(0.1032)	(0.0962)	(0.0962)
Pro-Gun Control $\times$ Black	0.5612	0.6118*	0.1359	0.1480
	(0.3455)	(0.3019)	(0.2817)	(0.2815)
Pro-Gun Control $\times$ Other	-0.2224	0.0455	0.1371	0.1687
	(0.2966)	(0.2604)	(0.2419)	(0.2428)
White	-0.5017*	-0.3822*	0.0284	0.0310
	(0.2188)	(0.1919)	(0.1784)	(0.1790)
Black	-0.3631	-0.3071	0.0019	-0.0232
	(0.2914)	(0.2559)	(0.2376)	(0.2386)
Controls	No	Yes	No	Yes
R ²	0.01297	0.25174	0.00941	0.01800
Observations	1,109	1,109	1,109	1,109

Notes: Pooled Sample. OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: gender, educational attainment, political ideology, and degree of political engagement.

Table 15: Heterogeneous Treatment Effects by Gender (Experiment 1)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.500*	6.506*	2.839*	2.841*
	(0.1313)	(0.1851)	(0.1121)	(0.1849)
Anti-Gun Control $\times$ Female	-0.2214	-0.2995*	0.1129	0.1001
	(0.1641)	(0.1409)	(0.1401)	(0.1408)
Anti-Gun Control $\times$ Male	-0.4840*	-0.4186*	0.0330	0.0554
	(0.1830)	(0.1573)	(0.1563)	(0.1571)
Pro-Gun Control $\times$ Female	0.2373	0.0798	0.1616	0.1341
	(0.1600)	(0.1380)	(0.1366)	(0.1378)
Pro-Gun Control $\times$ Male	0.0913	0.1585	0.1523	0.1674
	(0.1869)	(0.1601)	(0.1595)	(0.1599)
Female	0.3035	0.3565*	0.1899	0.1809
	(0.1703)	(0.1462)	(0.1454)	(0.1460)
Controls	No	Yes	No	Yes
R ²	0.04622	0.31065	0.01074	0.02161
Observations	812	812	812	812

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, educational attainment, political ideology, and degree of political engagement.

Table 16: Heterogeneous Treatment Effects by Gender (Experiment 2)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.188*	6.222*	2.925*	3.127*
	(0.1057)	(0.1610)	(0.0869)	(0.1502)
Anti-Gun Control $\times$ Female	0.0448	0.0354	-0.1086	-0.0951
	(0.1525)	(0.1347)	(0.1254)	(0.1256)
Anti-Gun Control $\times$ Male	-0.2834	-0.1834	-0.1205	-0.1248
	(0.1489)	(0.1313)	(0.1225)	(0.1224)
Pro-Gun Control $\times$ Female	0.2246	0.3640*	0.0800	0.0806
	(0.1528)	(0.1347)	(0.1256)	(0.1257)
Pro-Gun Control $\times$ Male	0.0595	0.1392	0.1419	0.1299
	(0.1452)	(0.1280)	(0.1194)	(0.1194)
Female	0.2580	0.2334	0.1112	0.1056
	(0.1478)	(0.1306)	(0.1215)	(0.1218)
Controls	No	Yes	No	Yes
R ²	0.02754	0.25542	0.00773	0.02237
Observations	1,109	1,109	1,109	1,109

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, educational attainment, political ideology, and degree of political engagement.

Table 17: Heterogeneous Treatment Effects by Educational Attainment (Experiment 1)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.500*	6.489*	3.043*	2.906*
	(0.1500)	(0.1935)	(0.1276)	(0.1935)
Anti-Gun Control $\times$ Post-Grad	-0.2498	-0.0874	0.1104	0.1164
	(0.2781)	(0.2365)	(0.2366)	(0.2364)
Anti-Gun Control $\times$ College	-0.3698*	-0.4021*	0.0682	0.0801
	(0.1730)	(0.1475)	(0.1471)	(0.1475)
Anti-Gun Control $\times$ HS/Below	-0.4867*	-0.4603*	-0.0168	0.0213
	(0.2238)	(0.1904)	(0.1904)	(0.1903)
Pro-Gun Control $\times$ Post-Grad	0.2963	0.3463	0.2222	0.2467
	(0.2768)	(0.2352)	(0.2355)	(0.2352)
Pro-Gun Control $\times$ College	0.0614	-0.0711	0.2257	0.2138
	(0.1760)	(0.1497)	(0.1497)	(0.1497)
Pro-Gun Control $\times$ HS/Below	0.2294	0.2124	0.0036	-0.0046
	(0.2164)	(0.1842)	(0.1841)	(0.1841)
Post-Grad	0.2593	0.0348	-0.1916	-0.2024
	(0.2466)	(0.2112)	(0.2098)	(0.2111)
College	0.2651	0.2347	-0.1039	-0.0861
	(0.1907)	(0.1627)	(0.1622)	(0.1627)
Controls	No	Yes	No	Yes
R ²	0.03298	0.31294	0.00457	0.02284
Observations	812	812	812	812

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, political ideology, and degree of political engagement.

Table 18: Heterogeneous Treatment Effects by Educational Attainment (Experiment 2)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	4.986*	6.158*	3.107*	3.227*
	(0.1218)	(0.1708)	(0.1001)	(0.1590)
Anti-Gun Control $\times$ Post-Grad	-0.2128	-0.0049	0.0517	0.1016
	(0.2644)	(0.2337)	(0.2173)	(0.2176)
Anti-Gun Control $\times$ College	-0.2763	-0.1158	-0.1339	-0.1349
	(0.1497)	(0.1323)	(0.1230)	(0.1232)
Anti-Gun Control $\times$ HS/Below	0.0328	-0.0551	-0.2183	-0.2192
	(0.1846)	(0.1629)	(0.1518)	(0.1517)
Pro-Gun Control $\times$ Post-Grad	-0.3519	0.0047	0.2000	0.2410
	(0.2621)	(0.2326)	(0.2154)	(0.2166)
Pro-Gun Control $\times$ College	0.1246	0.3031*	0.1738	0.1688
	(0.1503)	(0.1329)	(0.1235)	(0.1237)
Pro-Gun Control $\times$ HS/Below	0.2408	0.2340	-0.0603	-0.0638
	(0.1763)	(0.1556)	(0.1449)	(0.1448)
Post-Grad	0.8995*	0.6121*	-0.1071	-0.1226
	(0.2212)	(0.1963)	(0.1818)	(0.1828)
College	0.4143*	0.1778	-0.2044	-0.2046
	(0.1615)	(0.1431)	(0.1327)	(0.1333)
Controls	No	Yes	No	Yes
R ²	0.03051	0.25496	0.01156	0.02511
Observations	1,109	1,109	1,109	1,109

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, political ideology, and degree of political engagement.

Table 19: HTEs by Selection Propensity (Experiment 1, Passed Attention Check)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.634*	6.444*	3.010*	2.814*
	(0.1457)	(0.2056)	(0.1224)	(0.2043)
Anti-Gun Control $\times$ High Propensity	-0.2817	-0.4393*	0.0829	0.1478
	(0.2289)	(0.1946)	(0.1922)	(0.1934)
Anti-Gun Control $\times$ Medium Propensity	-0.4787	-0.3999	0.2685	0.3345
	(0.2574)	(0.2174)	(0.2162)	(0.2160)
Anti-Gun Control $\times$ Low Propensity	-0.4798*	-0.4422*	0.0157	0.0468
	(0.2208)	(0.1861)	(0.1854)	(0.1849)
Pro-Gun Control $\times$ High Propensity	0.1768	0.2205	0.3085	0.3170
	(0.2120)	(0.1784)	(0.1780)	(0.1773)
Pro-Gun Control $\times$ Medium Propensity	0.0425	0.0445	0.1344	0.1704
	(0.2459)	(0.2074)	(0.2065)	(0.2061)
Pro-Gun Control $\times$ Low Propensity	0.0599	-0.1642	-0.0260	-0.0341
	(0.2363)	(0.1990)	(0.1985)	(0.1978)
High Propensity	0.1333	0.1727	-0.0487	-0.0703
	(0.2051)	(0.1731)	(0.1723)	(0.1720)
Medium Propensity	0.0910	0.1207	-0.2128	-0.2193
	(0.2288)	(0.1923)	(0.1921)	(0.1912)
Controls	No	Yes	No	Yes
R ²	0.02674	0.32532	0.00955	0.03832
Observations	704	704	704	704

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, political ideology, and degree of political engagement. This table replicates [Figure 2](#) on the subset of participants who pass both attention checks regarding the content of the ad that they watch.

Table 20: HTEs by Skipping Early (Experiment 2)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.291*	6.130*	2.996*	3.164*
	(0.0889)	(0.1606)	(0.0728)	(0.1499)
Anti-Gun Control $\times$ Skipped Early	0.1423	0.1143	-0.1698	-0.1727
	(0.1793)	(0.1573)	(0.1468)	(0.1468)
Anti-Gun Control $\times$ Watched	-0.3686*	-0.2133	-0.0743	-0.0544
	(0.1347)	(0.1187)	(0.1103)	(0.1108)
Pro-Gun Control $\times$ Skipped Early	0.1351	0.2712	0.0902	0.1042
	(0.1849)	(0.1623)	(0.1514)	(0.1515)
Pro-Gun Control $\times$ Watched	0.0947	0.2341*	0.1185	0.1153
	(0.1285)	(0.1129)	(0.1052)	(0.1054)
Skipped Early	0.0974	-0.0013	-0.0479	-0.0601
	(0.1612)	(0.1421)	(0.1320)	(0.1326)
Controls	No	Yes	No	Yes
R ²	0.01958	0.25799	0.00755	0.02435
Observations	1,109	1,109	1,109	1,109

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, political ideology, and degree of political engagement. This table computes heterogeneous treatment effects for the set of participants who tried to skip the ad before 60 seconds had passed.

Table 21: Robustness to Removing Idlers (Experiment 2)

	Attitude		Issue Priority	
	(1)	(2)	(3)	(4)
Constant	5.373*	6.258*	2.913*	3.062*
	(0.1289)	(0.1790)	(0.1053)	(0.1670)
Anti-Gun Control $\times$ High Propensity	-0.2141	-0.0551	-0.1030	-0.0833
	(0.1888)	(0.1654)	(0.1543)	(0.1543)
Anti-Gun Control $\times$ Medium Propensity	0.1723	0.1231	-0.1154	-0.1403
	(0.1831)	(0.1602)	(0.1496)	(0.1494)
Anti-Gun Control $\times$ Low Propensity	-0.3292	-0.2511	-0.2109	-0.1762
	(0.1871)	(0.1645)	(0.1529)	(0.1535)
Pro-Gun Control $\times$ High Propensity	0.0768	0.2644	0.0805	0.0873
	(0.1842)	(0.1621)	(0.1505)	(0.1513)
Pro-Gun Control $\times$ Medium Propensity	0.2979	0.3961*	-0.1117	-0.1274
	(0.1848)	(0.1618)	(0.1510)	(0.1510)
Pro-Gun Control $\times$ Low Propensity	-0.0746	0.0790	0.3050*	0.3131*
	(0.1831)	(0.1605)	(0.1496)	(0.1498)
High Propensity	0.1184	-0.0661	0.1735	0.1798
	(0.1862)	(0.1632)	(0.1522)	(0.1523)
Medium Propensity	-0.2462	-0.2339	0.0575	0.0651
	(0.1796)	(0.1573)	(0.1467)	(0.1468)
Controls	No	Yes	No	Yes
R ²	0.01179	0.25515	0.01672	0.03351
Observations	1,095	1,095	1,095	1,095

Notes: OLS estimates. IID standard errors in parentheses. Significance: * $p < 0.05$. Controls: race, gender, educational attainment, political ideology, and degree of political engagement. This table computes heterogeneous treatment effects excluding the participants who idle for 90 seconds or more.